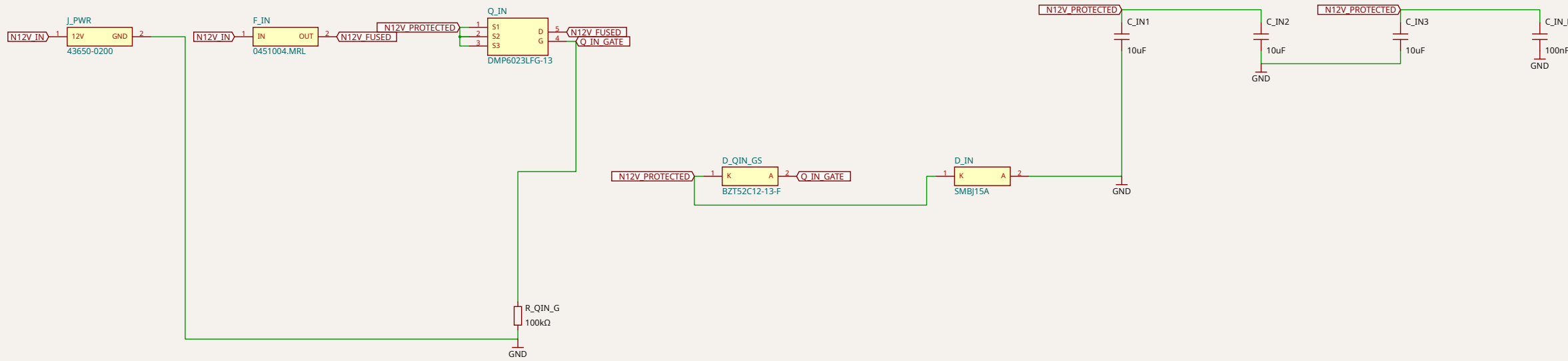


# Crow USB Carrier v1 — PROTOTYPE ONLY

## POWER INPUT

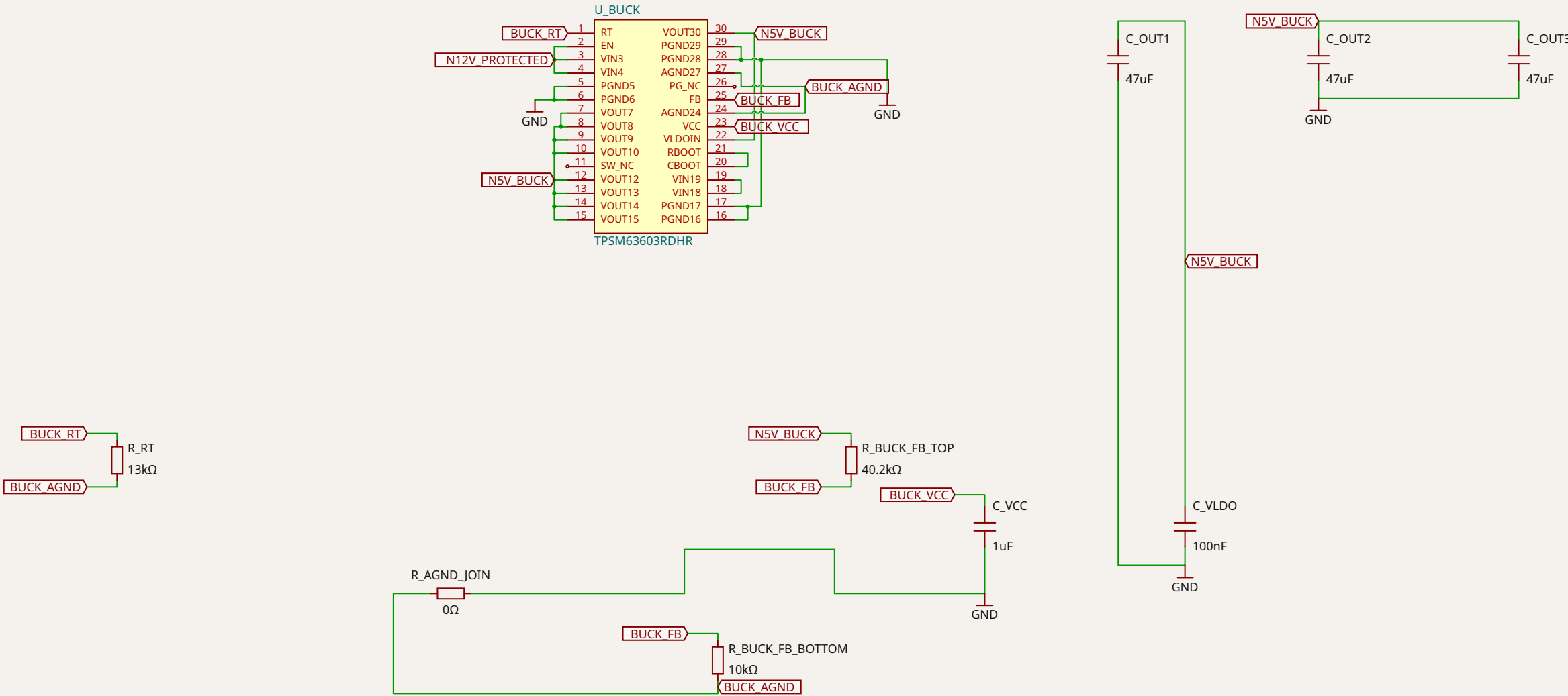
Page 1 of 45 • exact circuit.json SHA-256 580ac31b0de8d376... • 10 components • landscape page fit



# Crow USB Carrier v1 — PROTOTYPE ONLY

## POWER BUCK

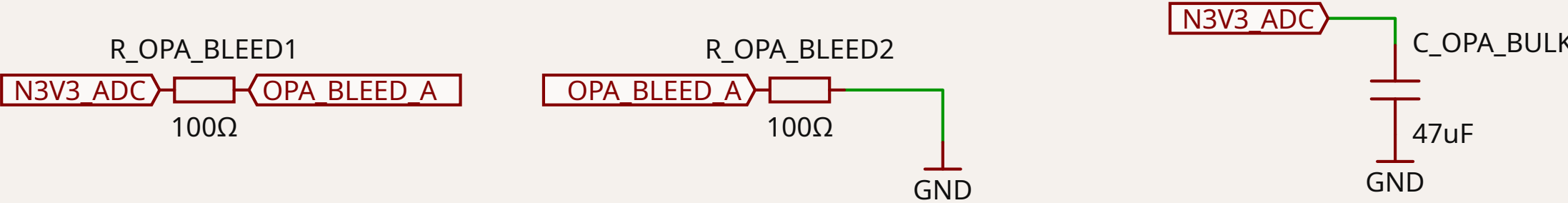
Page 2 of 45 • exact circuit.json SHA-256 580ac31b0de8d376... • 10 components • landscape page fit



# Crow USB Carrier v1 — PROTOTYPE ONLY

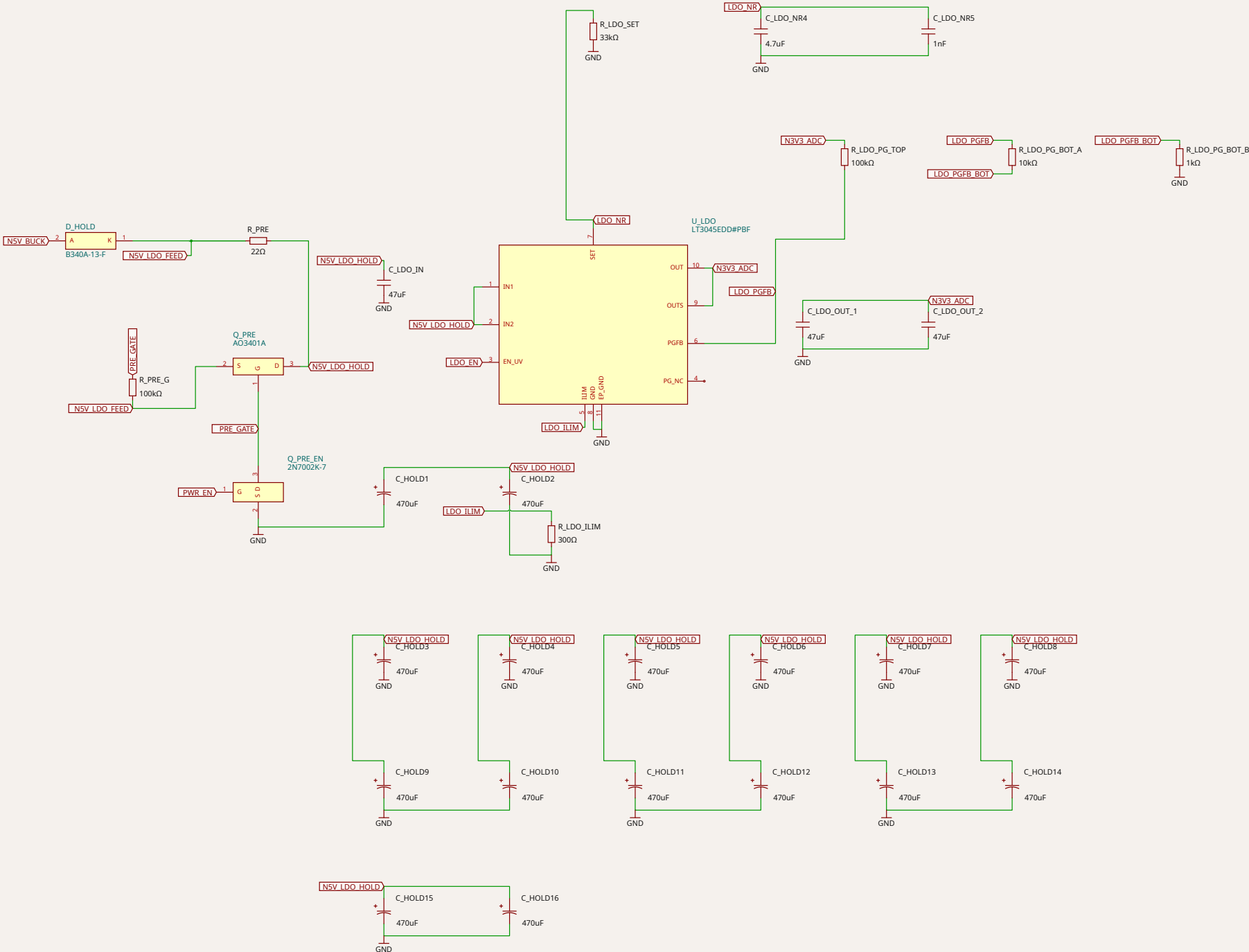
## ANALOG POWER AUX

Page 3 of 45 • exact circuit.json SHA-256 580ac31b0de8d376... • 3 components • landscape page fit



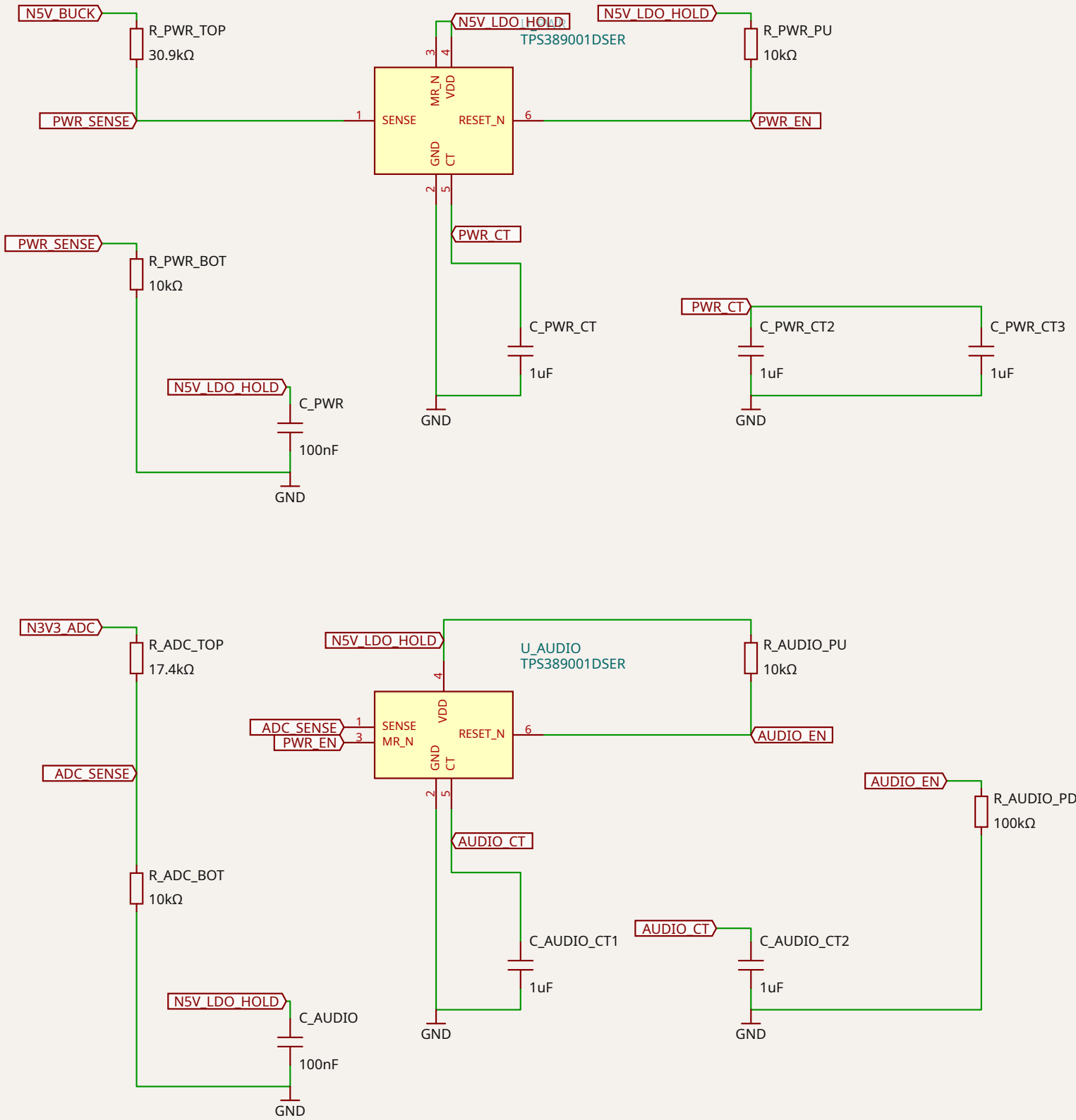
## Crow USB Carrier v1 — PROTOTYPE ONLY

Page 4 of 45 • exact circuit.json SHA-256 580ac31b0de8d376... • 32 components • landscape page fit



# Crow USB Carrier v1 — PROTOTYPE ONLY

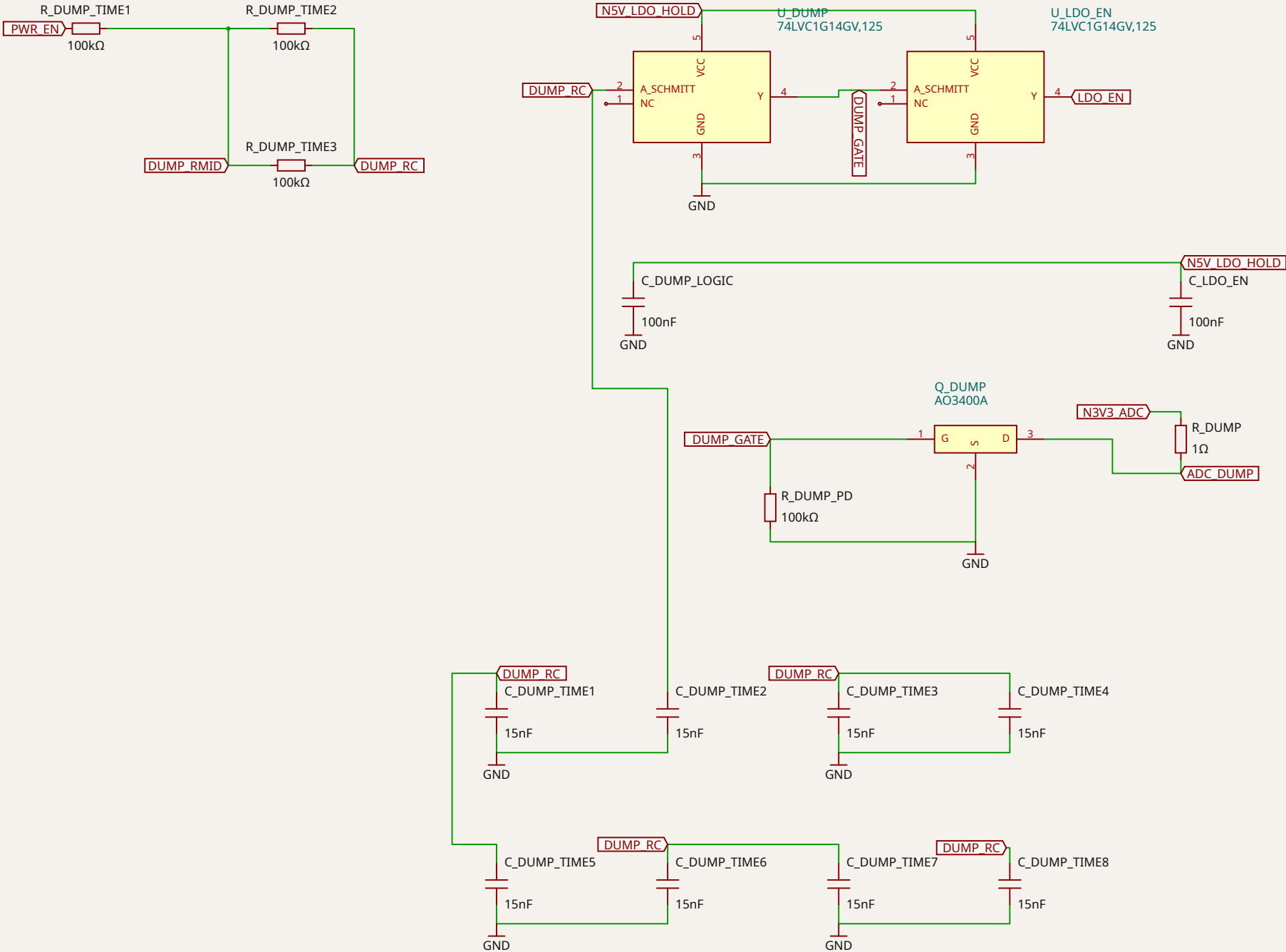
## SUPERVISORS



# Crow USB Carrier v1 — PROTOTYPE ONLY

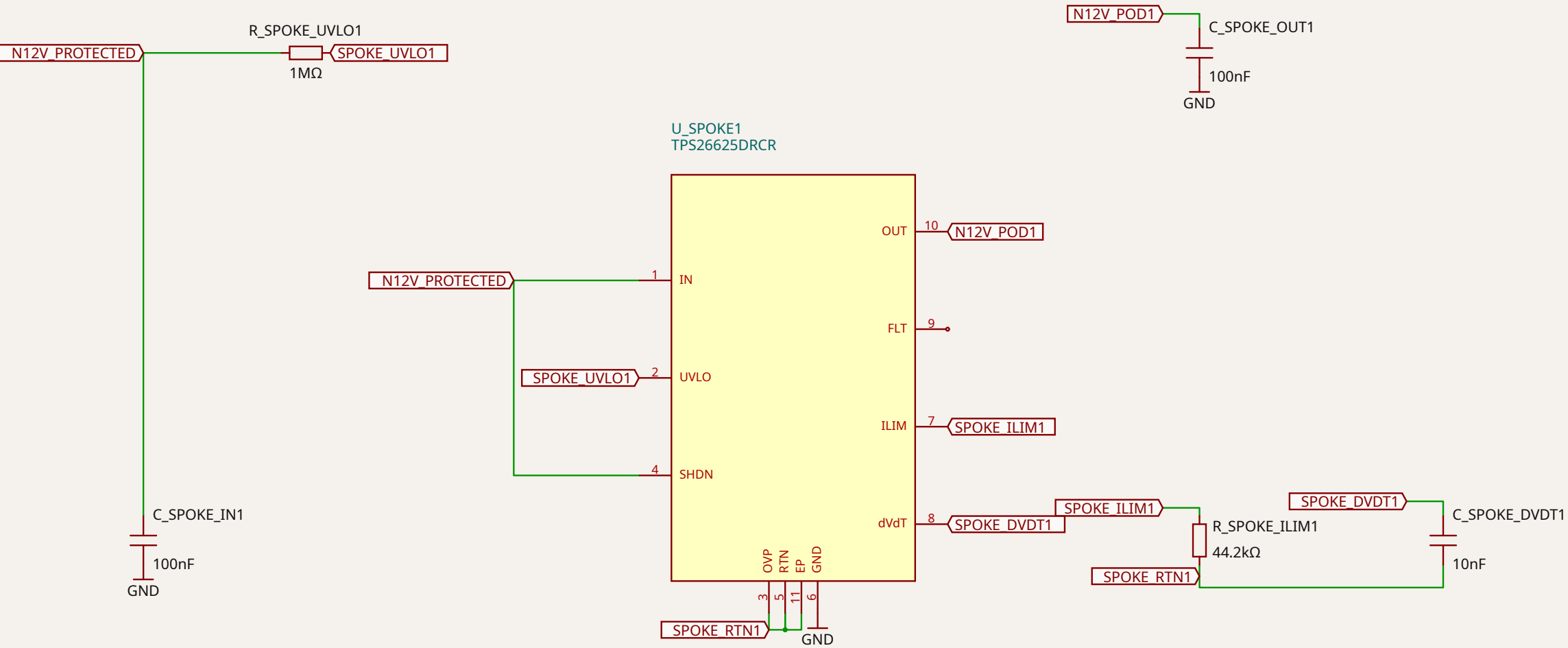
## DUMP

Page 6 of 45 • exact circuit.json SHA-256 580ac31b0de8d376... • 18 components • landscape page fit



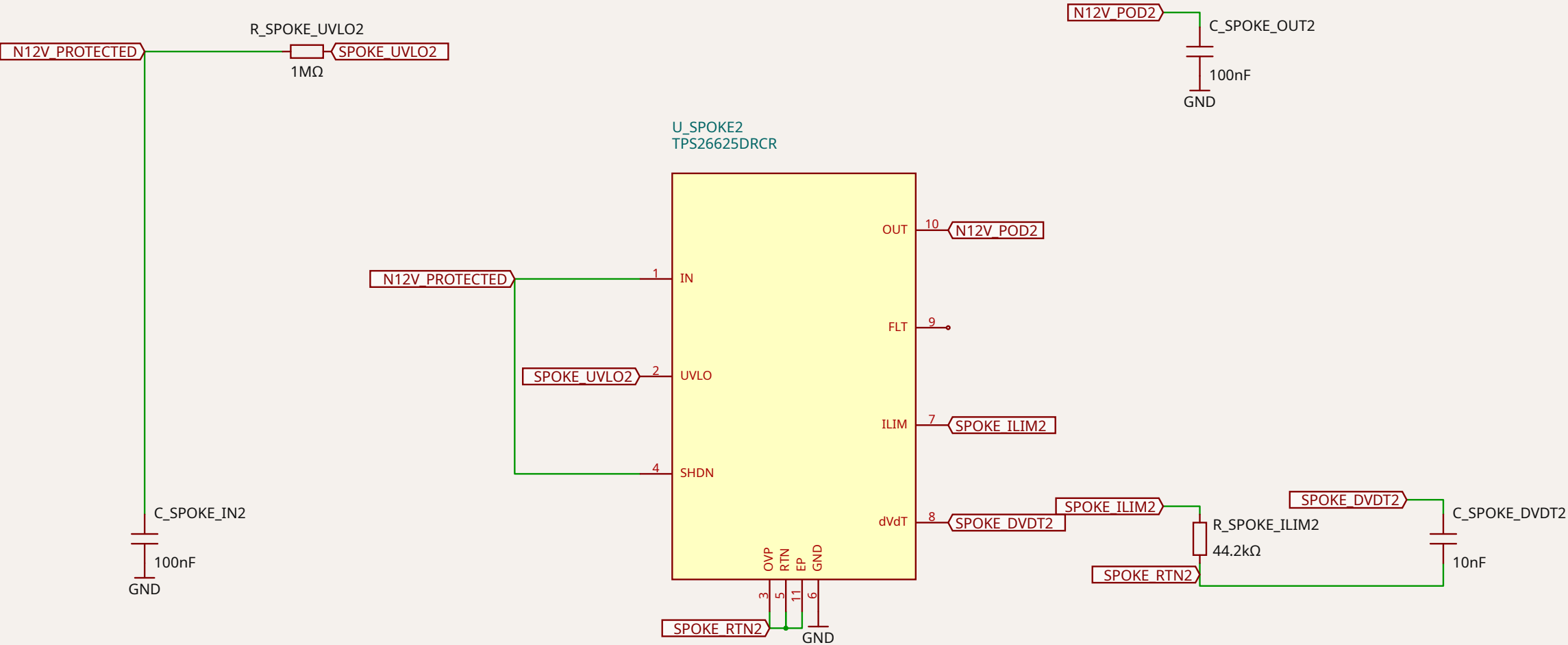
# Crow USB Carrier v1 — PROTOTYPE ONLY

## SPOKE PROTECTION 1



# Crow USB Carrier v1 — PROTOTYPE ONLY

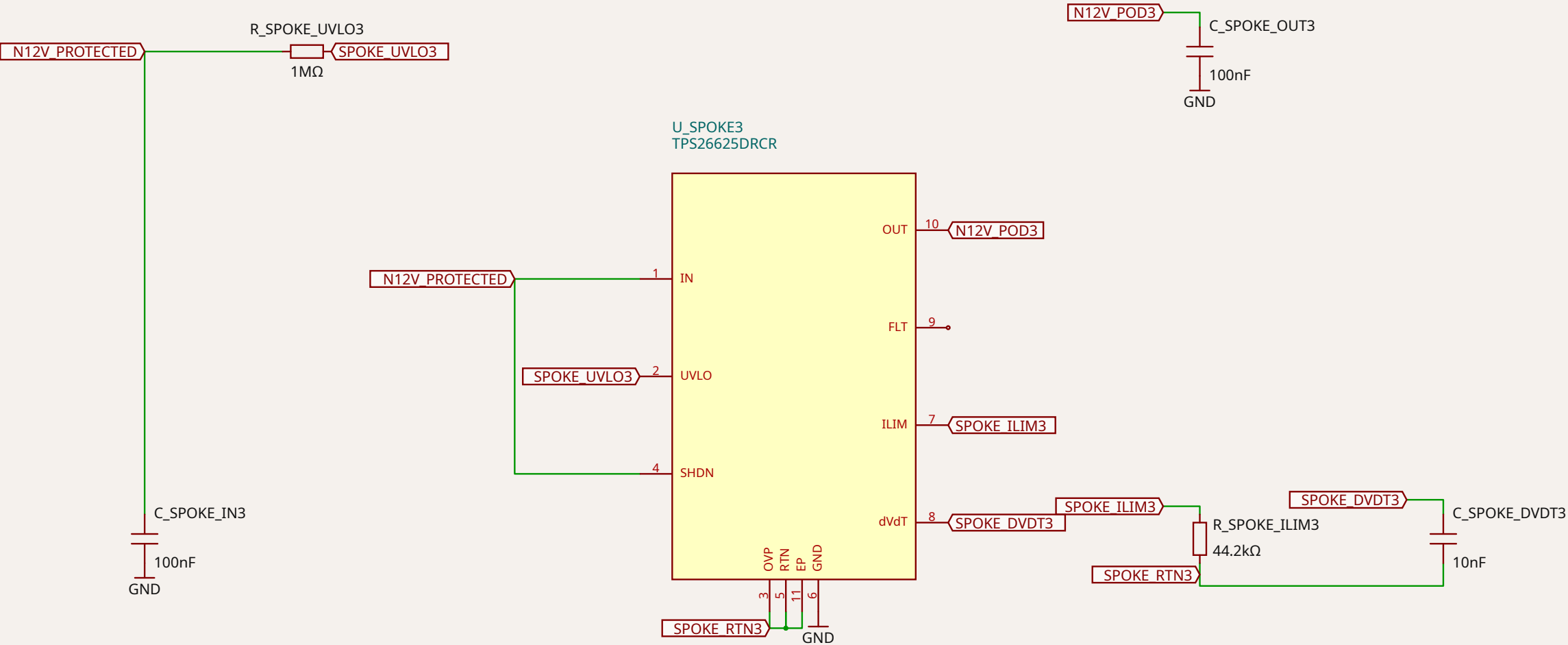
## SPOKE PROTECTION 2



# Crow USB Carrier v1 — PROTOTYPE ONLY

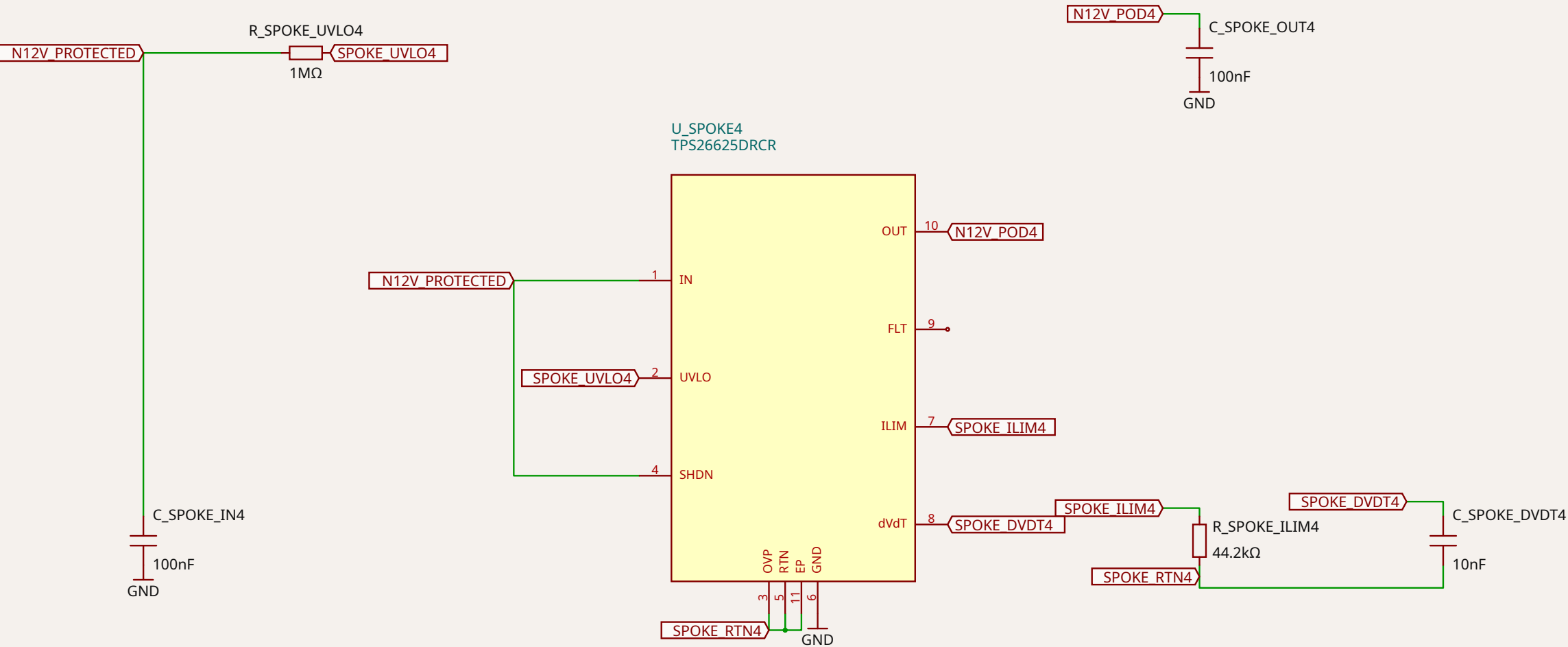
## SPOKE PROTECTION 3

Page 9 of 45 • exact circuit.json SHA-256 580ac31b0de8d376... • 6 components • landscape page fit



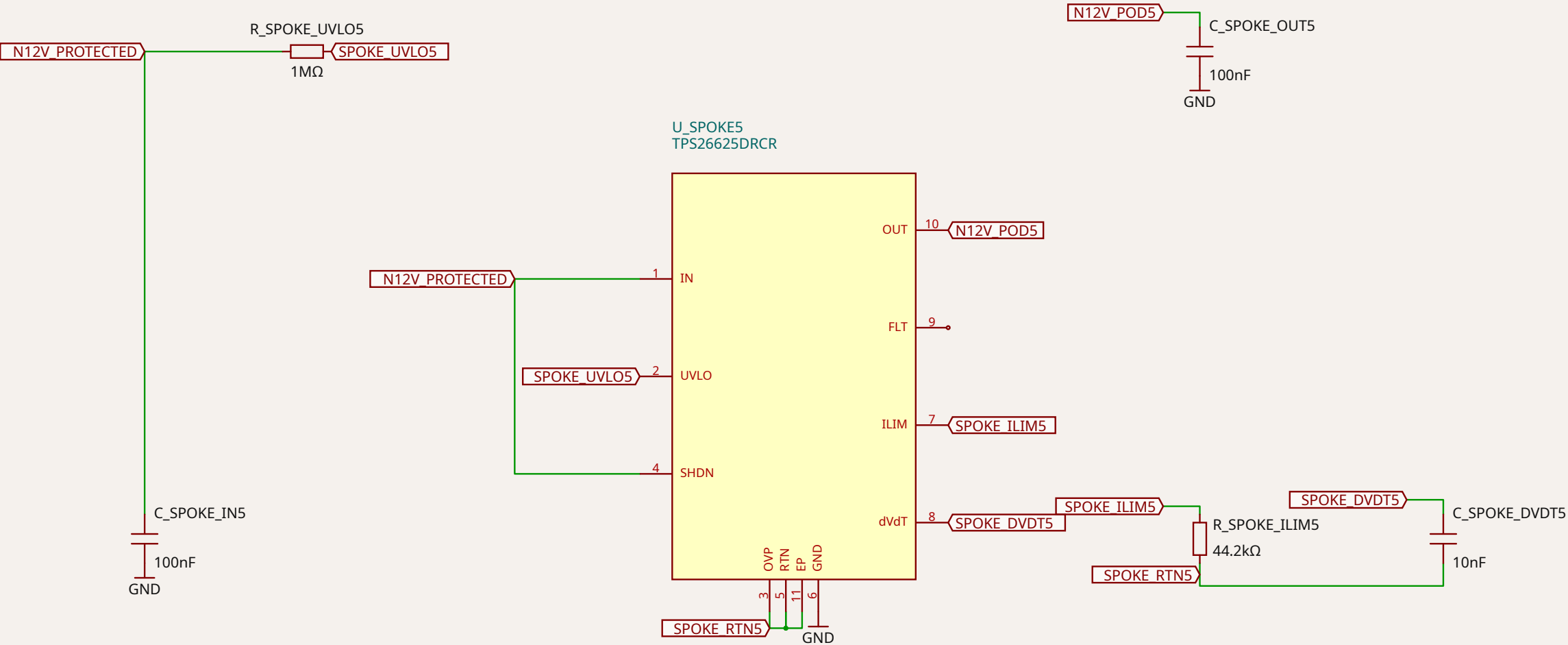
# Crow USB Carrier v1 — PROTOTYPE ONLY

## SPOKE PROTECTION 4



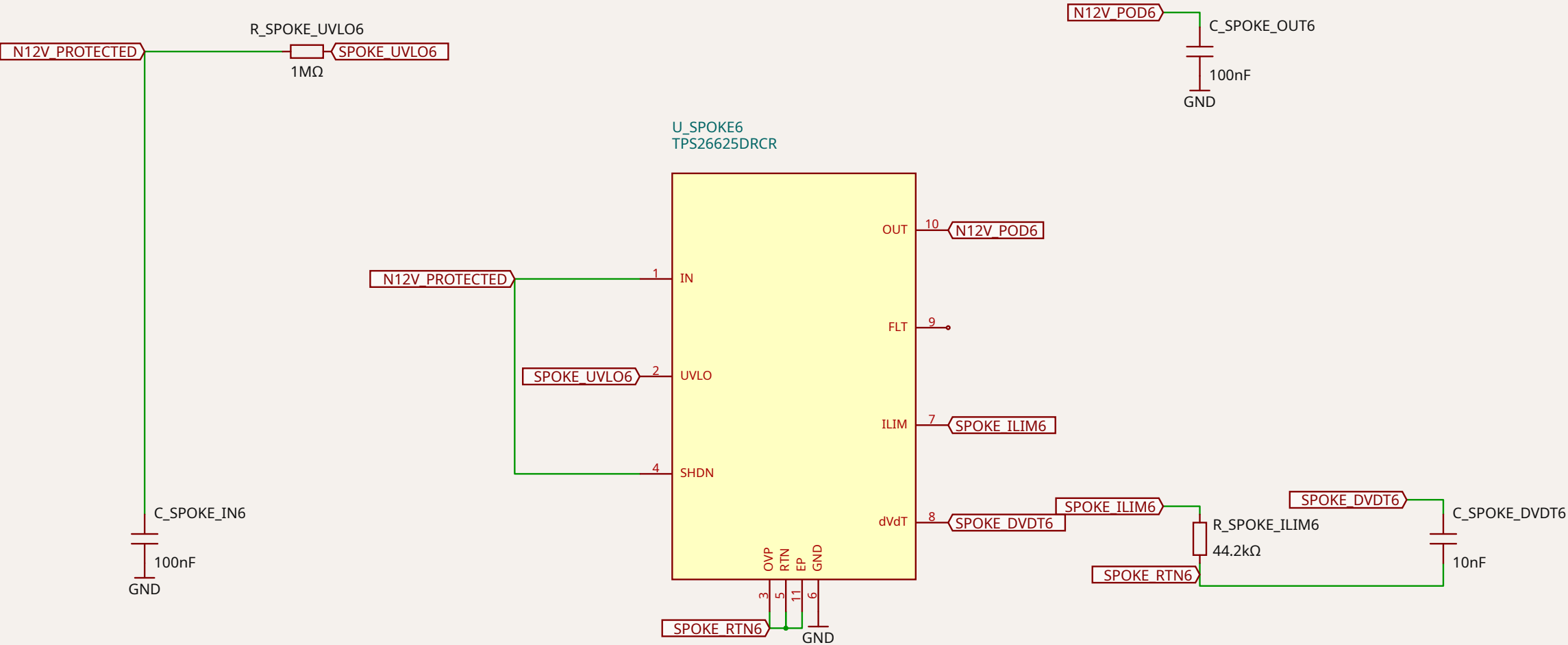
# Crow USB Carrier v1 — PROTOTYPE ONLY

## SPOKE PROTECTION 5



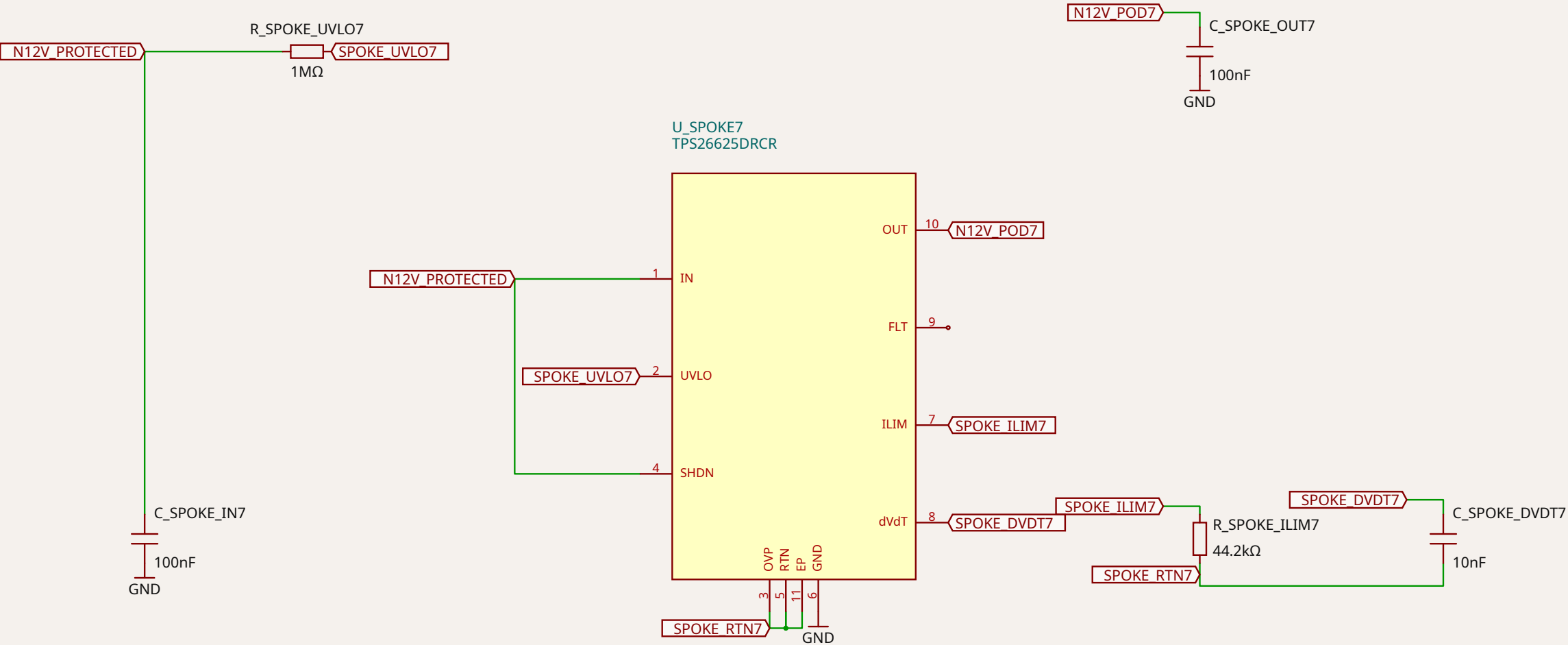
# Crow USB Carrier v1 — PROTOTYPE ONLY

## SPOKE PROTECTION 6



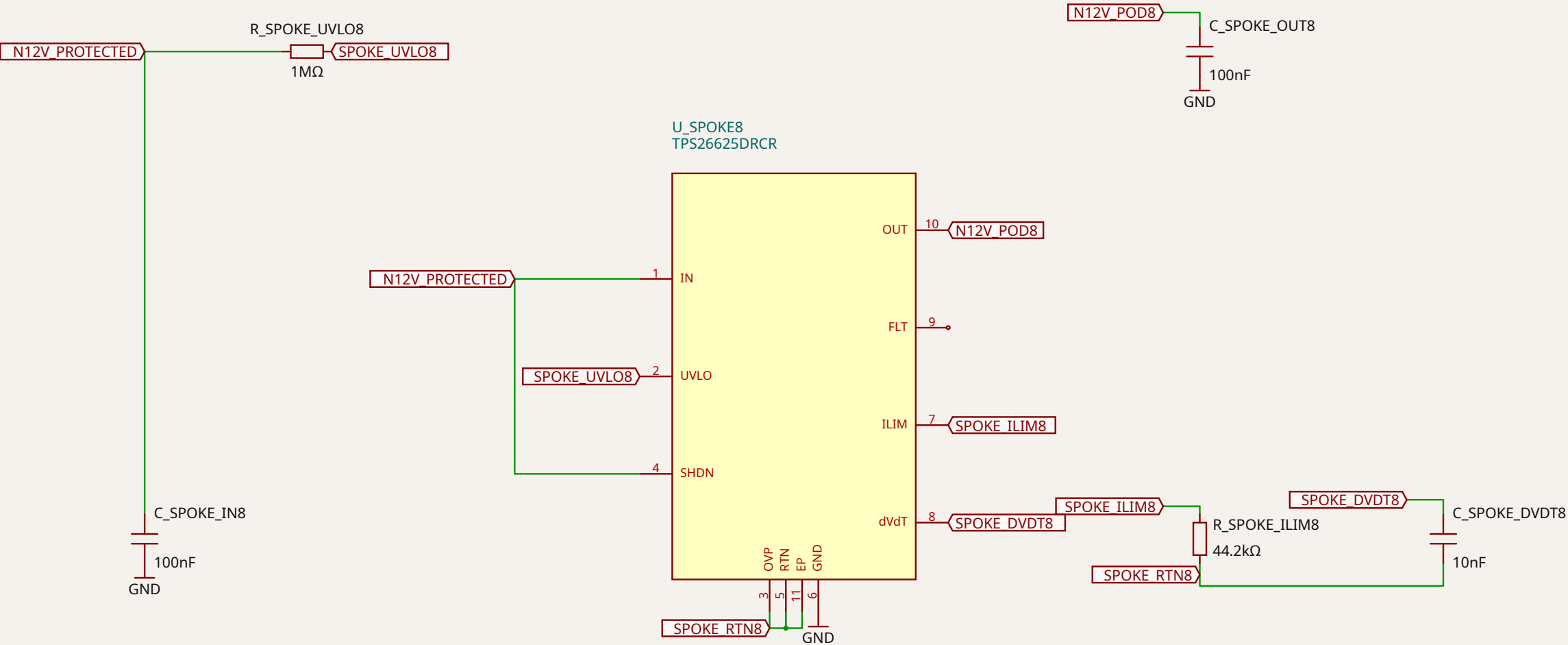
# Crow USB Carrier v1 — PROTOTYPE ONLY

## SPOKE PROTECTION 7



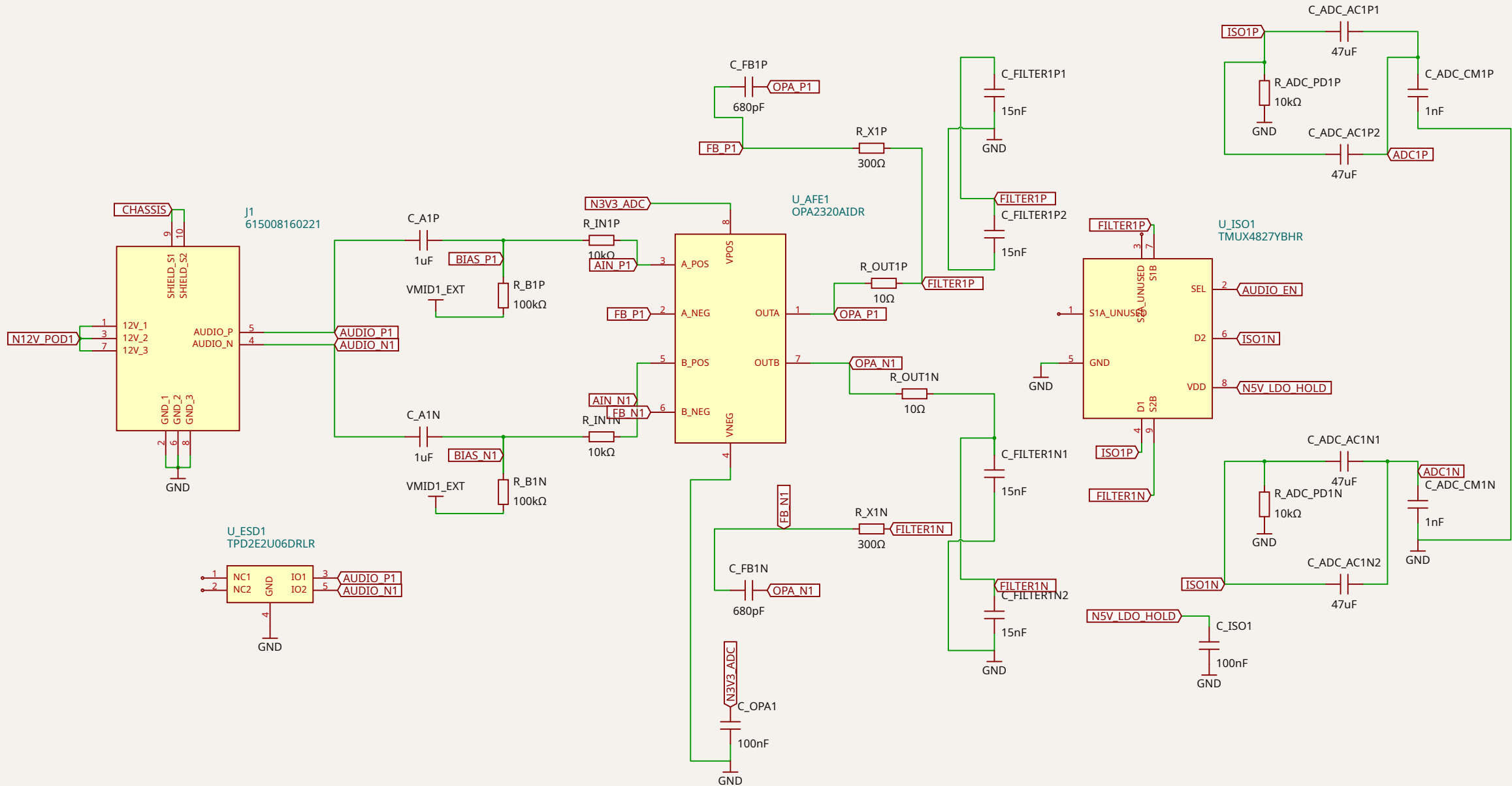
# Crow USB Carrier v1 — PROTOTYPE ONLY

## SPOKE PROTECTION 8



## ANALOG 1

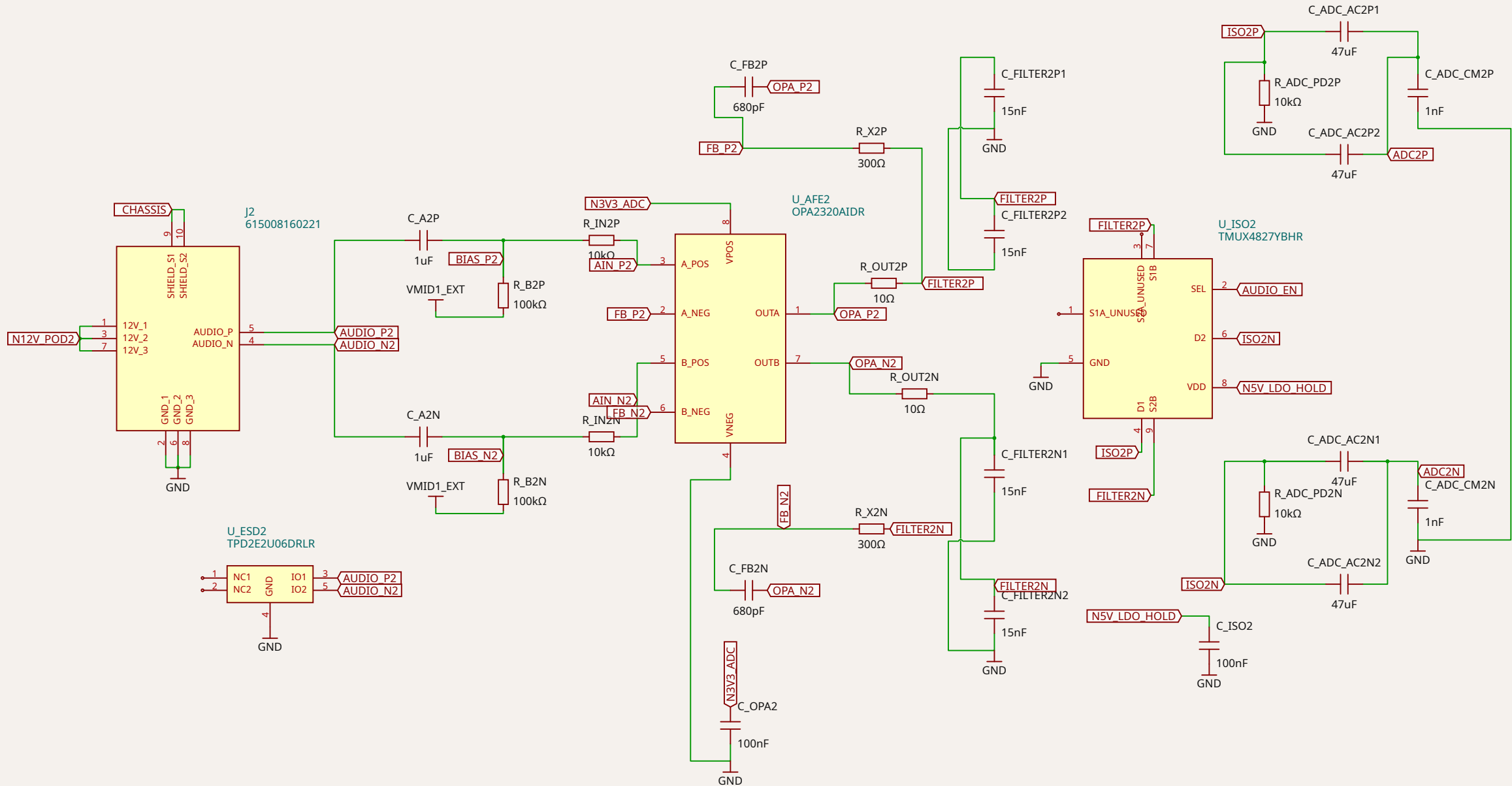
Page 15 of 45 • exact circuit.json SHA-256 580ac31b0de8d376... • 30 components • landscape page fit



## Crow USB Carrier v1 — PROTOTYPE ONLY

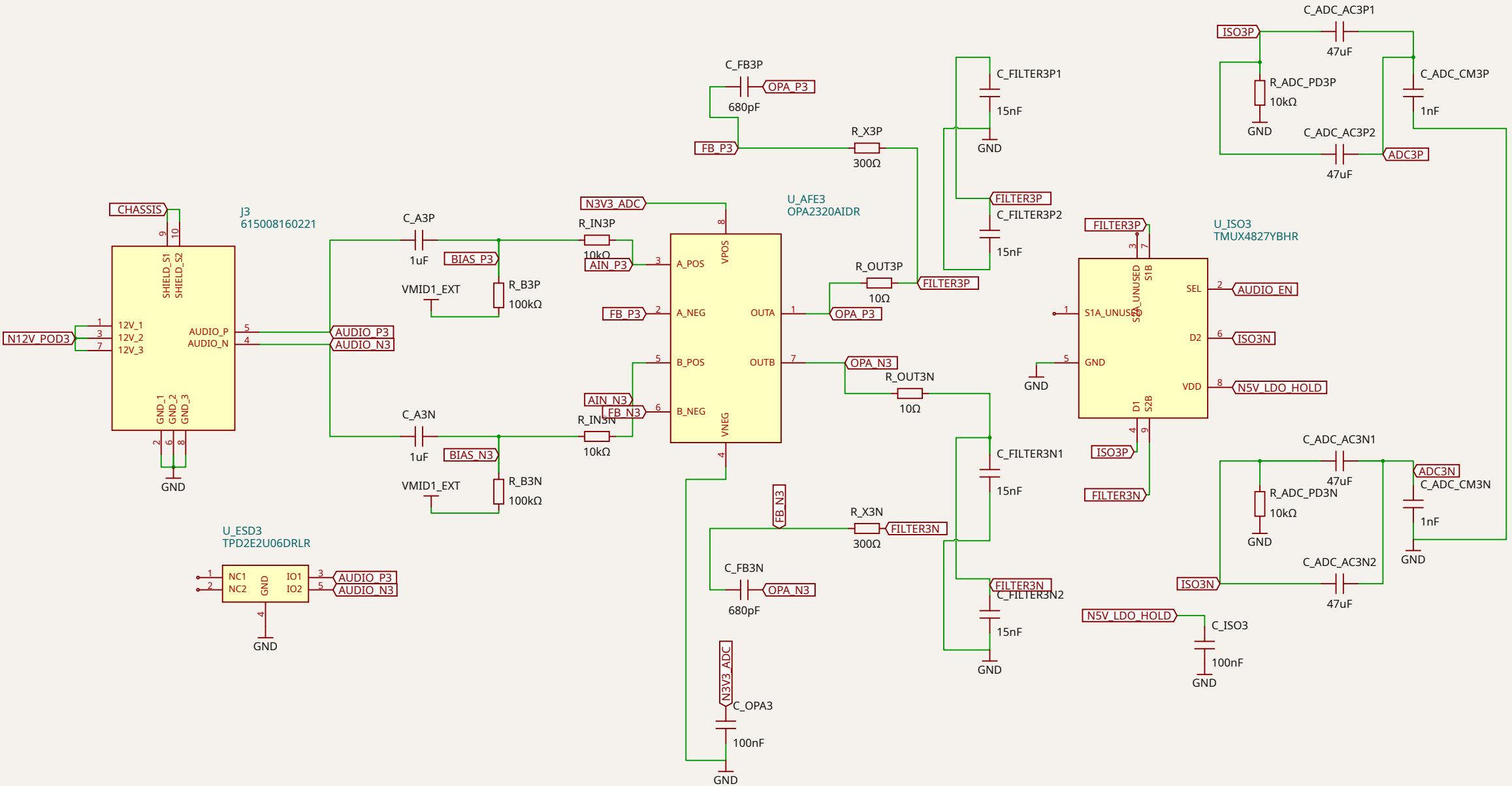
### ANALOG 2

Page 16 of 45 • exact circuit.json SHA-256 580ac31b0de8d376... • 30 components • landscape page fit

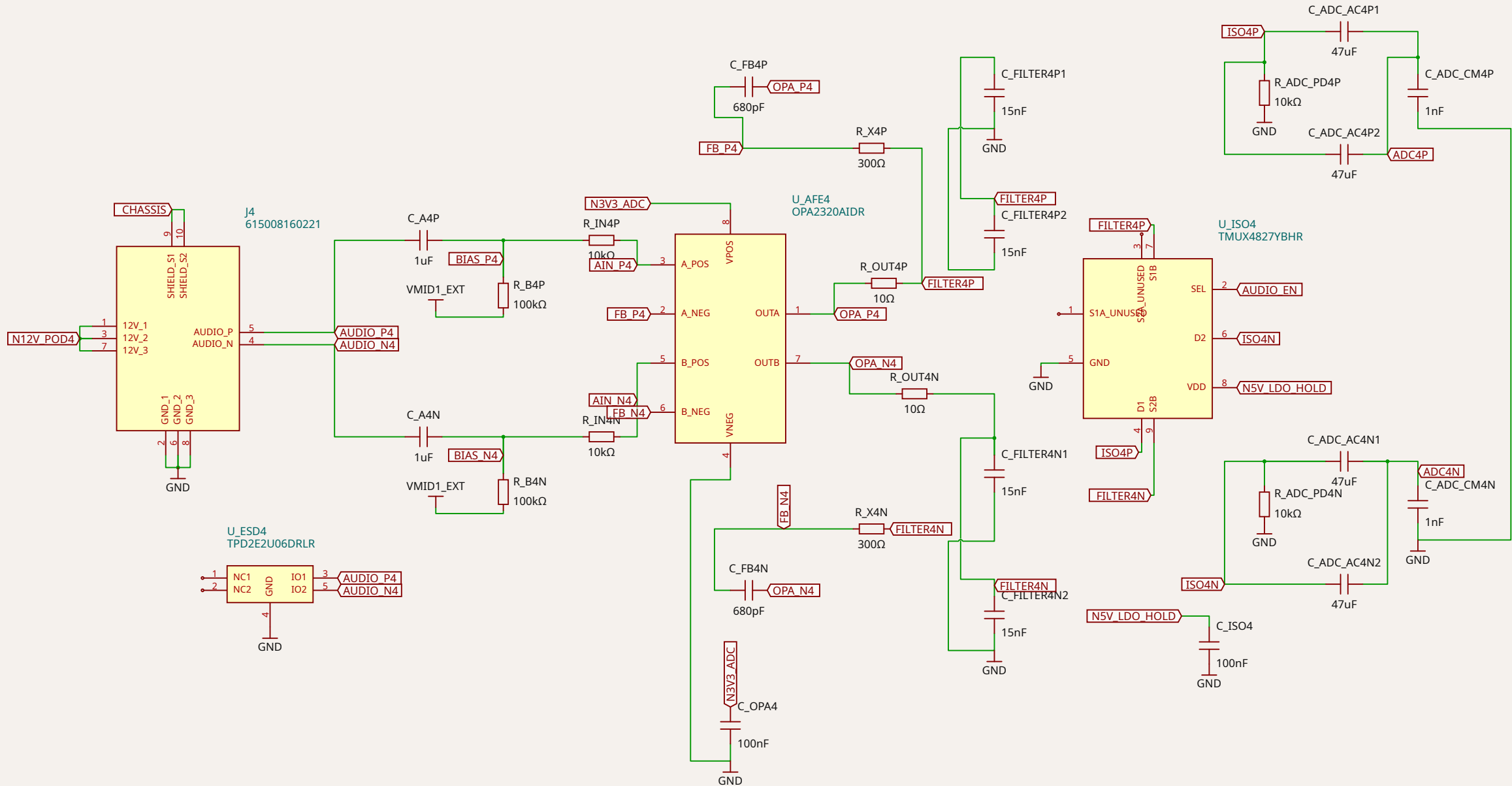


# Crow USB Carrier v1 — PROTOTYPE ONLY

## ANALOG 3

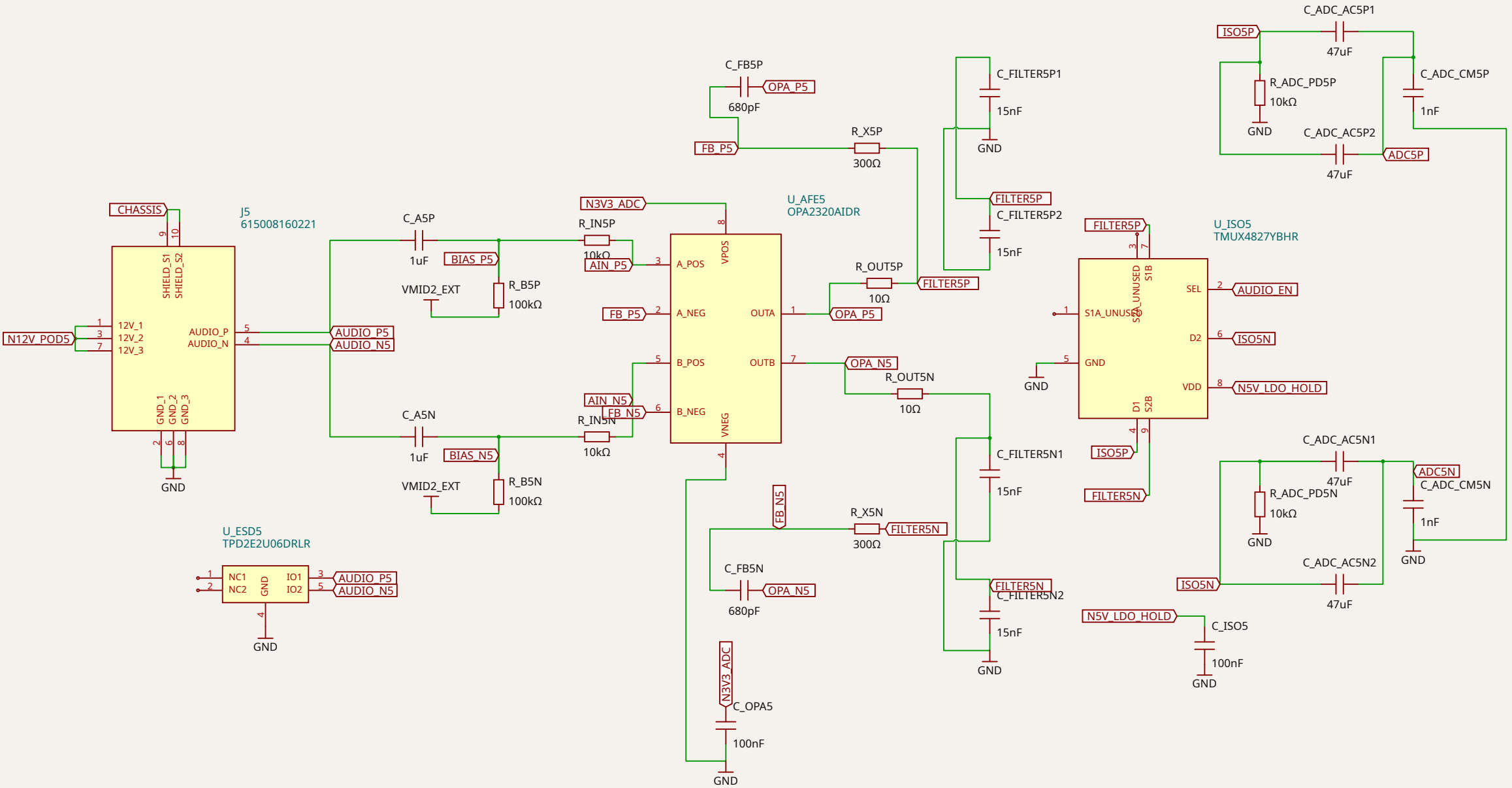


## ANALOG 4



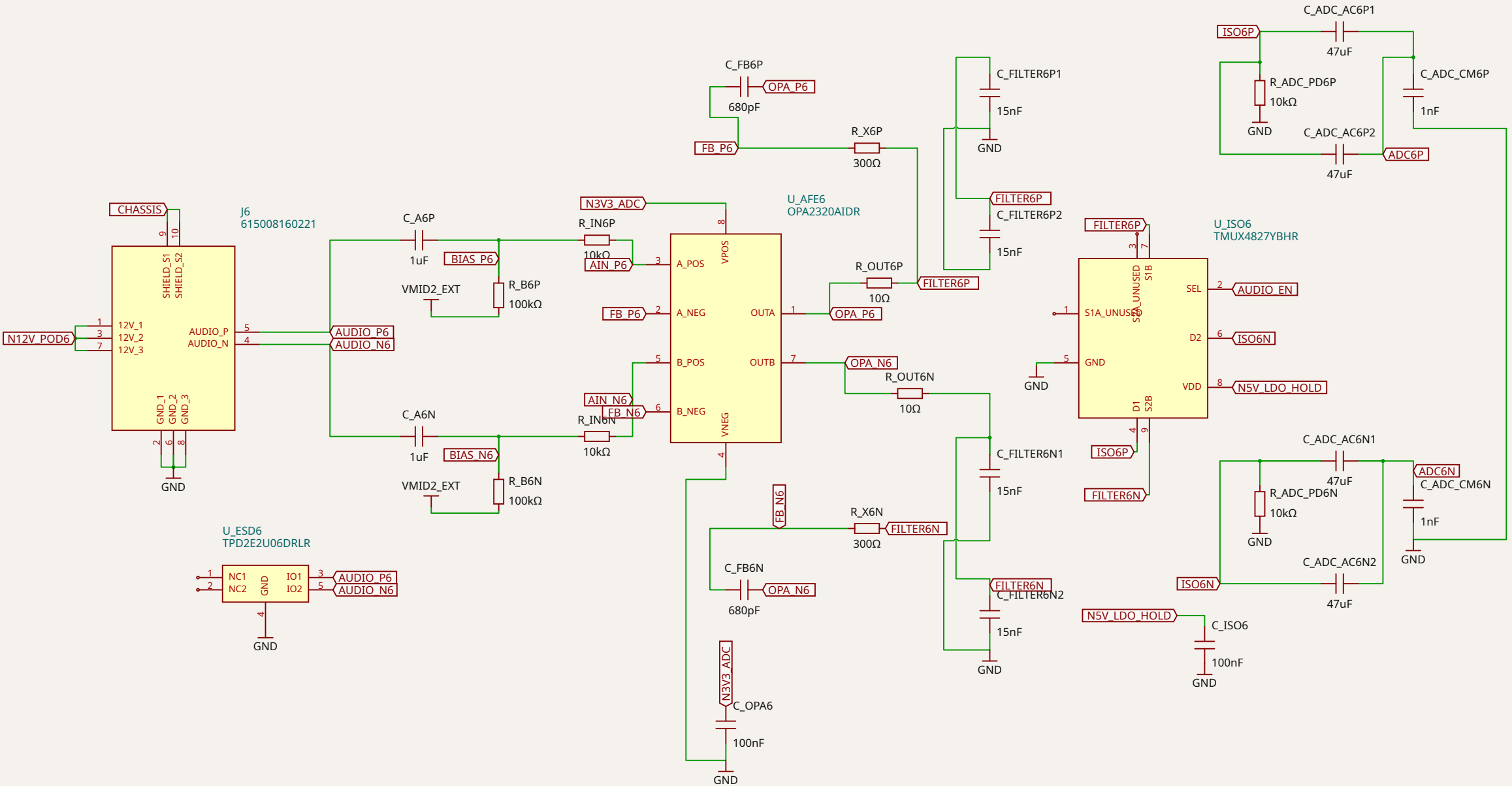
# Crow USB Carrier v1 — PROTOTYPE ONLY

## ANALOG 5



# Crow USB Carrier v1 — PROTOTYPE ONLY

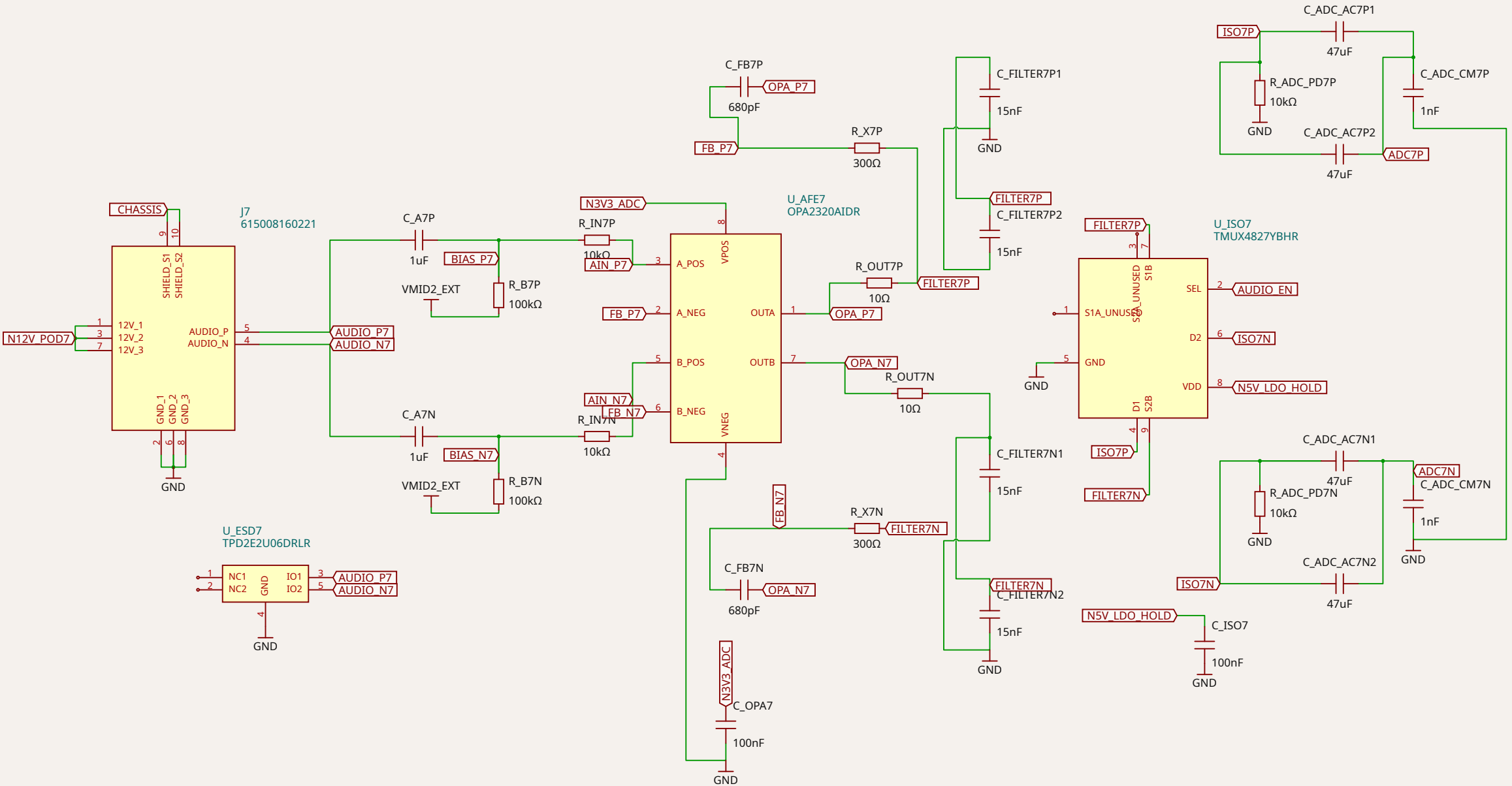
## ANALOG 6



# Crow USB Carrier v1 — PROTOTYPE ONLY

## ANALOG 7

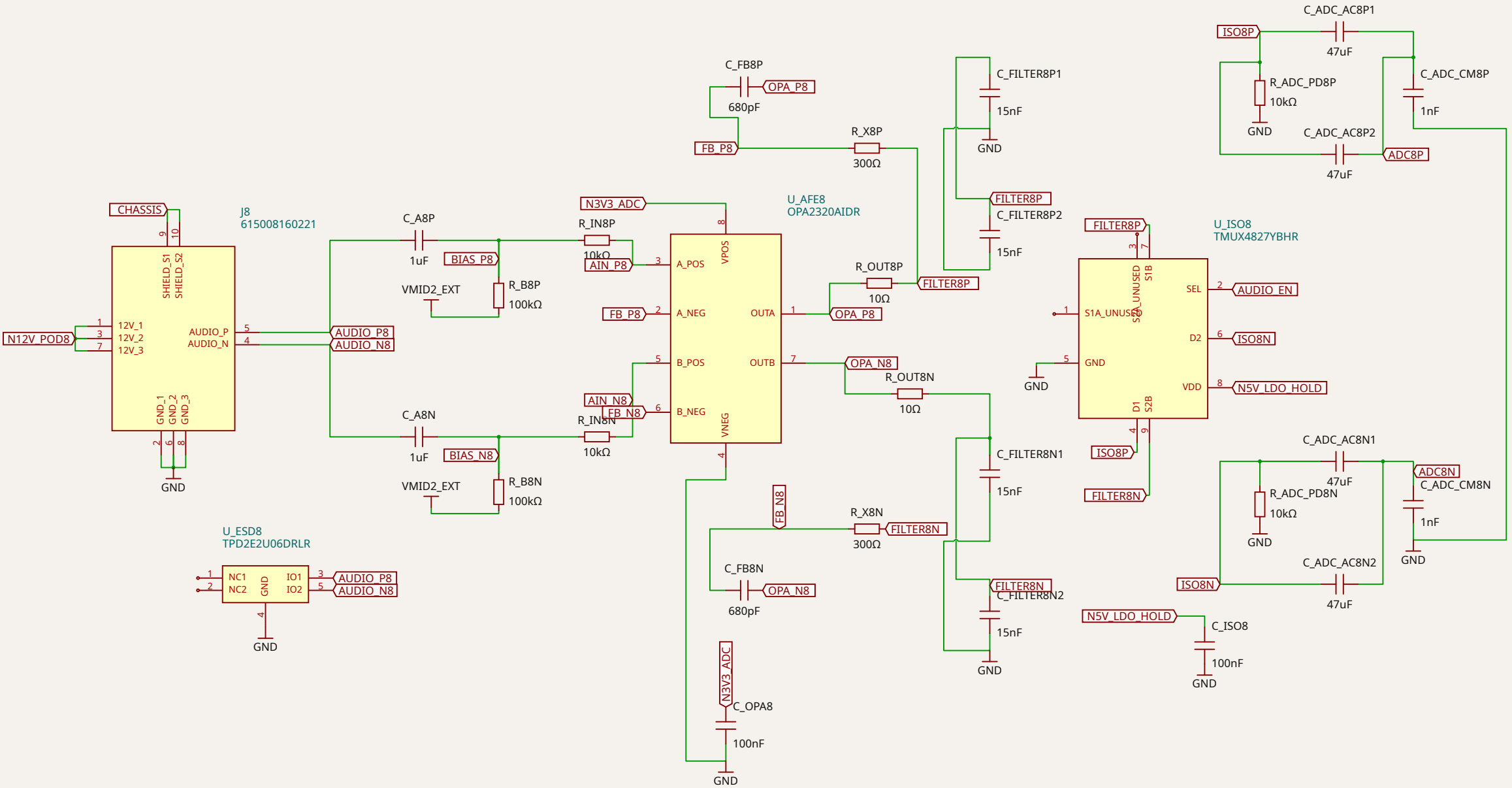
Page 21 of 45 • exact circuit.json SHA-256 580ac31b0de8d376... • 30 components • landscape page fit



# Crow USB Carrier v1 — PROTOTYPE ONLY

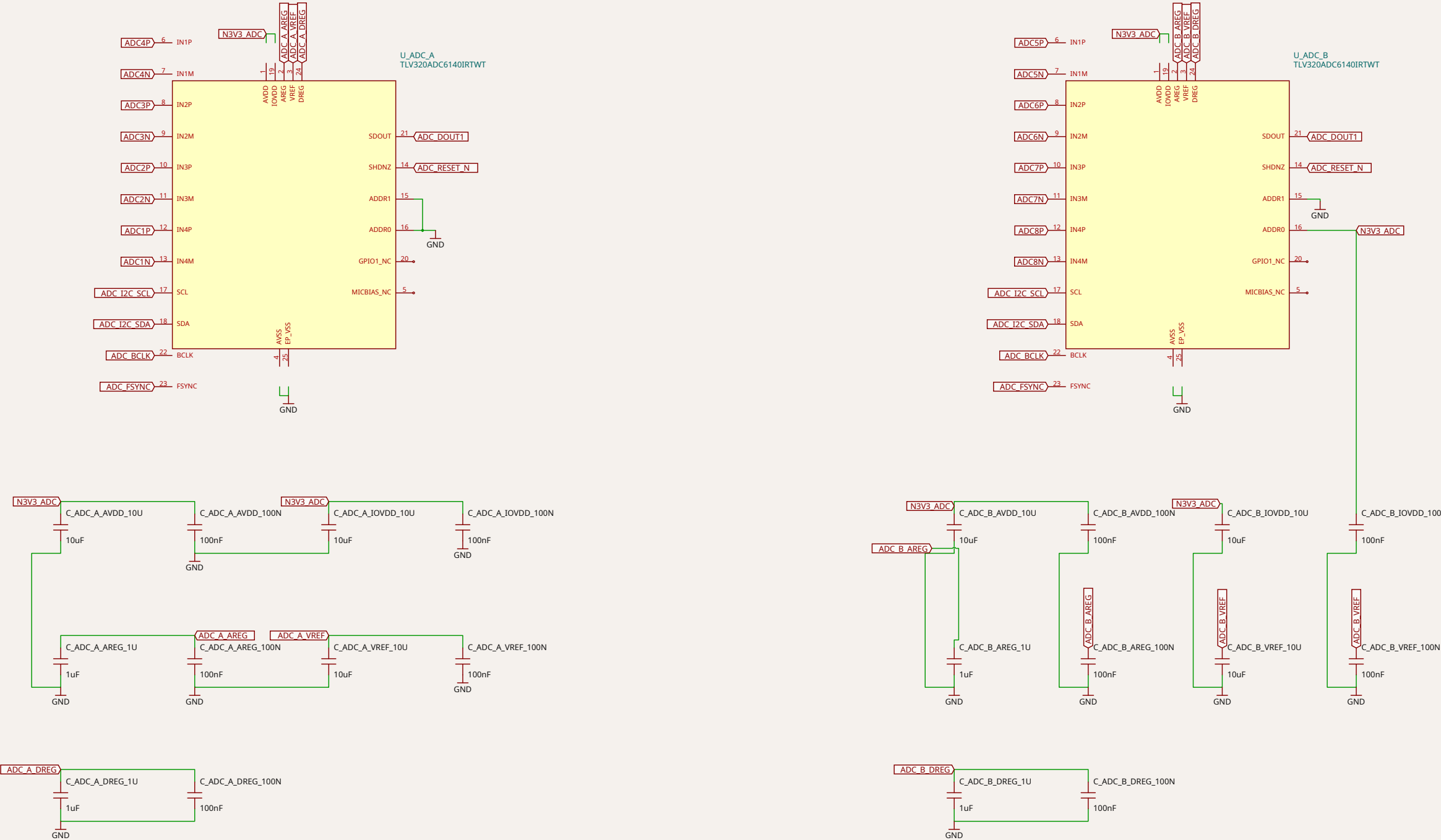
## ANALOG 8

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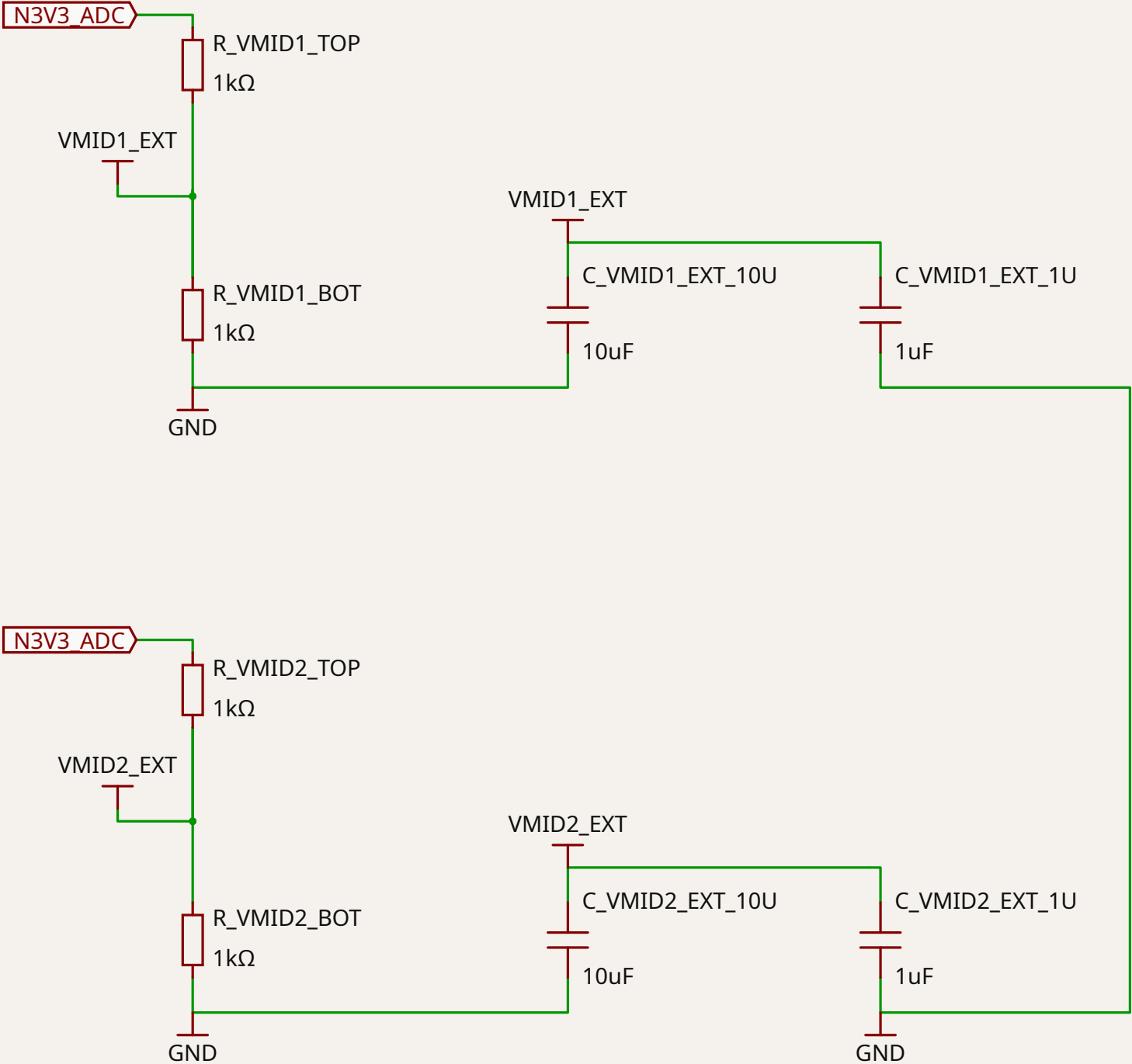
# Crow USB Carrier v1 — PROTOTYPE ONLY

## ADC



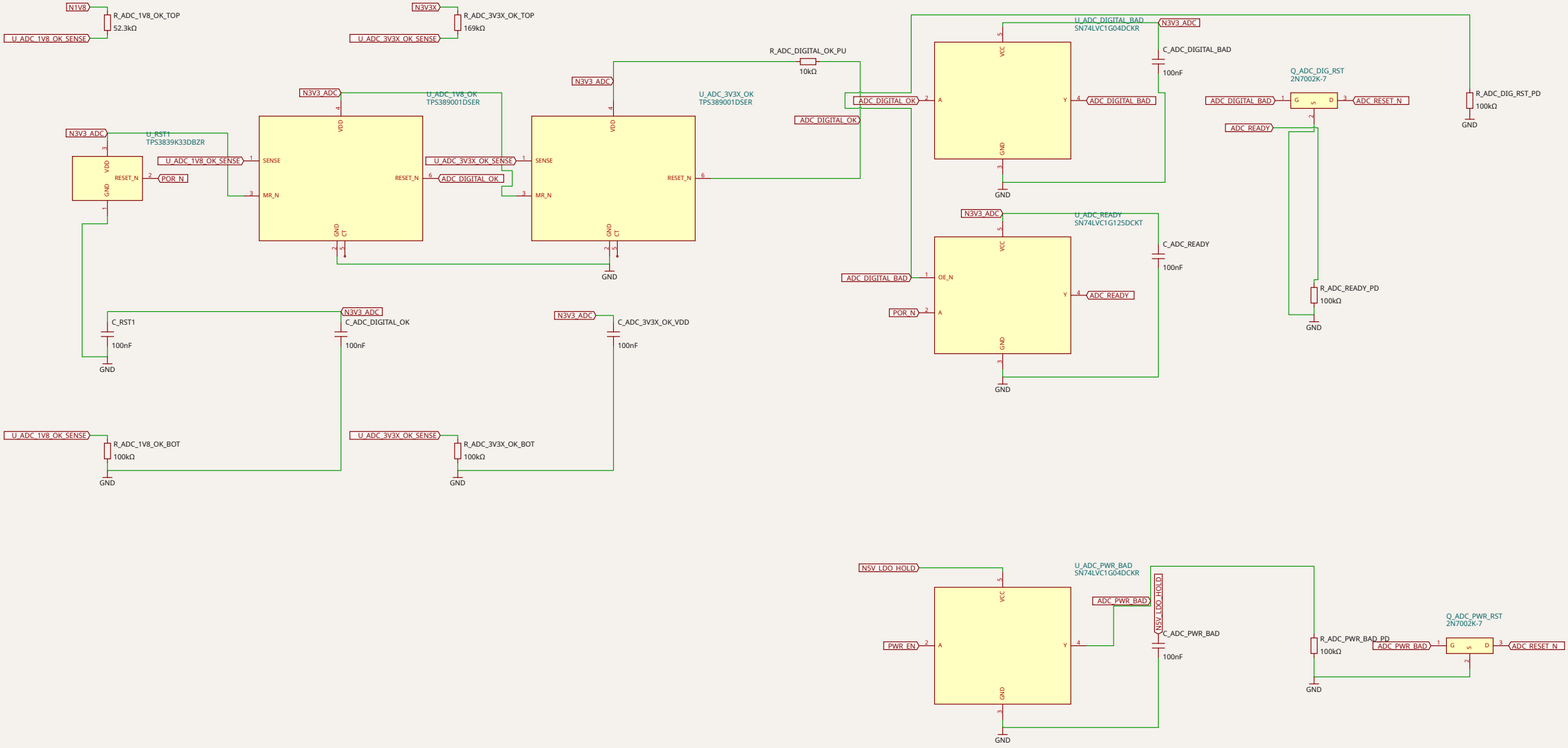
# Crow USB Carrier v1 — PROTOTYPE ONLY

## VMID



# Crow USB Carrier v1 — PROTOTYPE ONLY

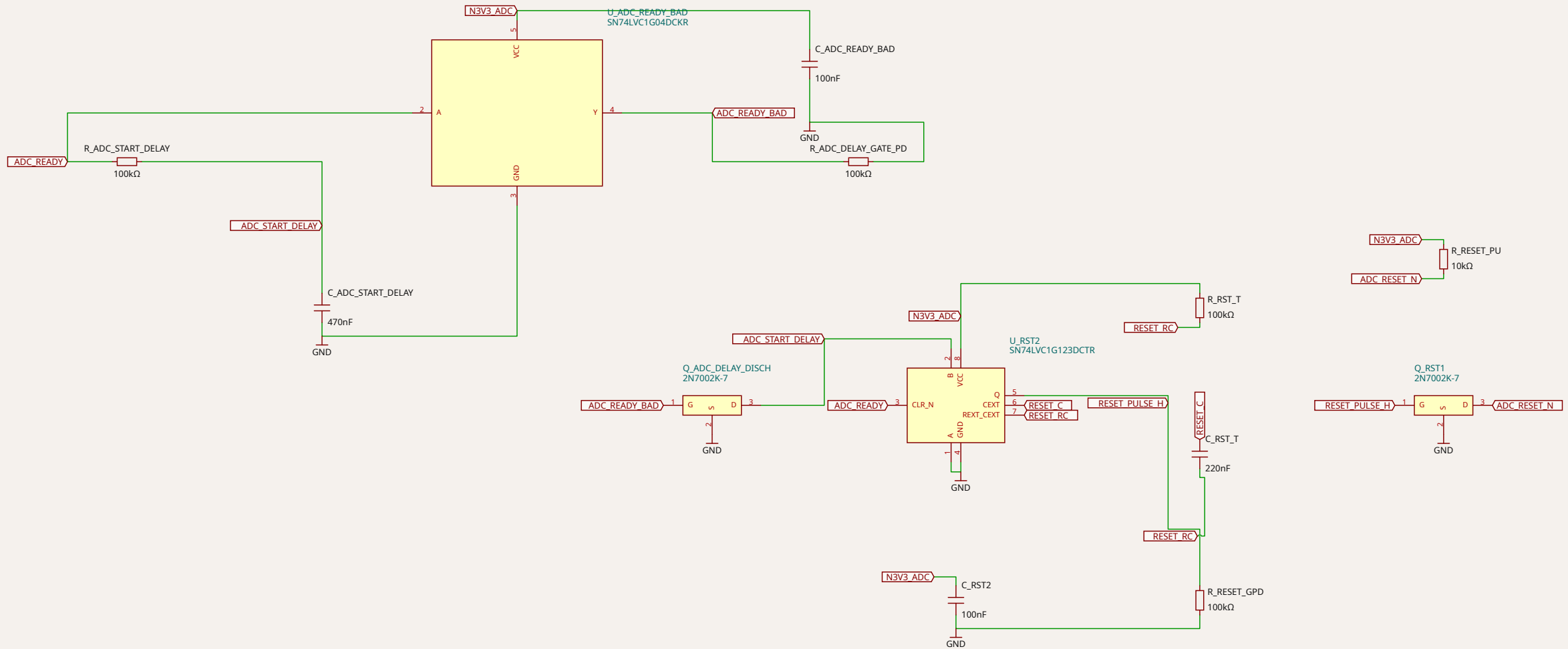
## RESET SUPERVISORS



## Crow USB Carrier v1 — PROTOTYPE ONLY

### RESET SEQUENCER

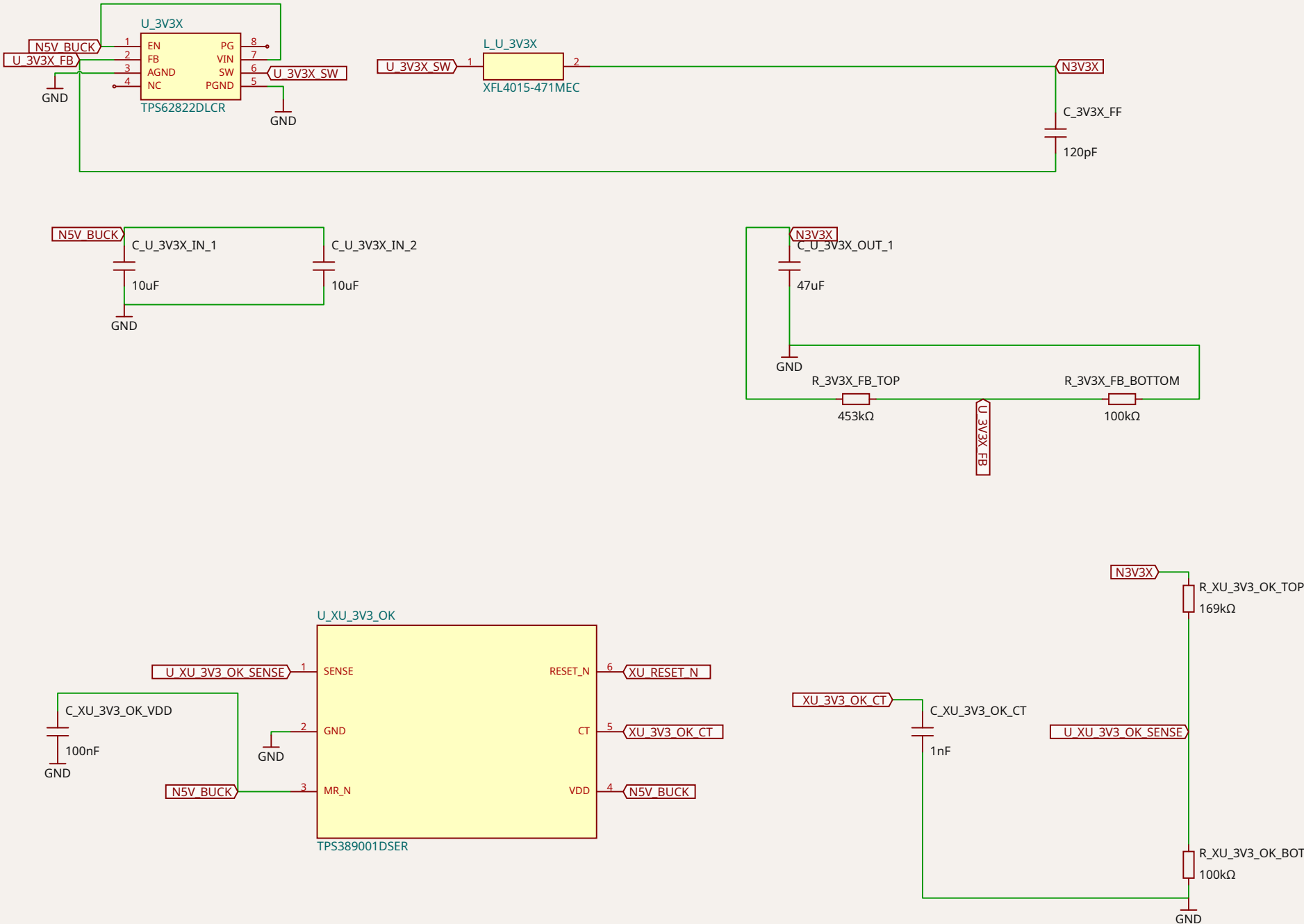
Page 26 of 45 • exact circuit.json SHA-256 580ac31b0de8d376... • 13 components • landscape page fit



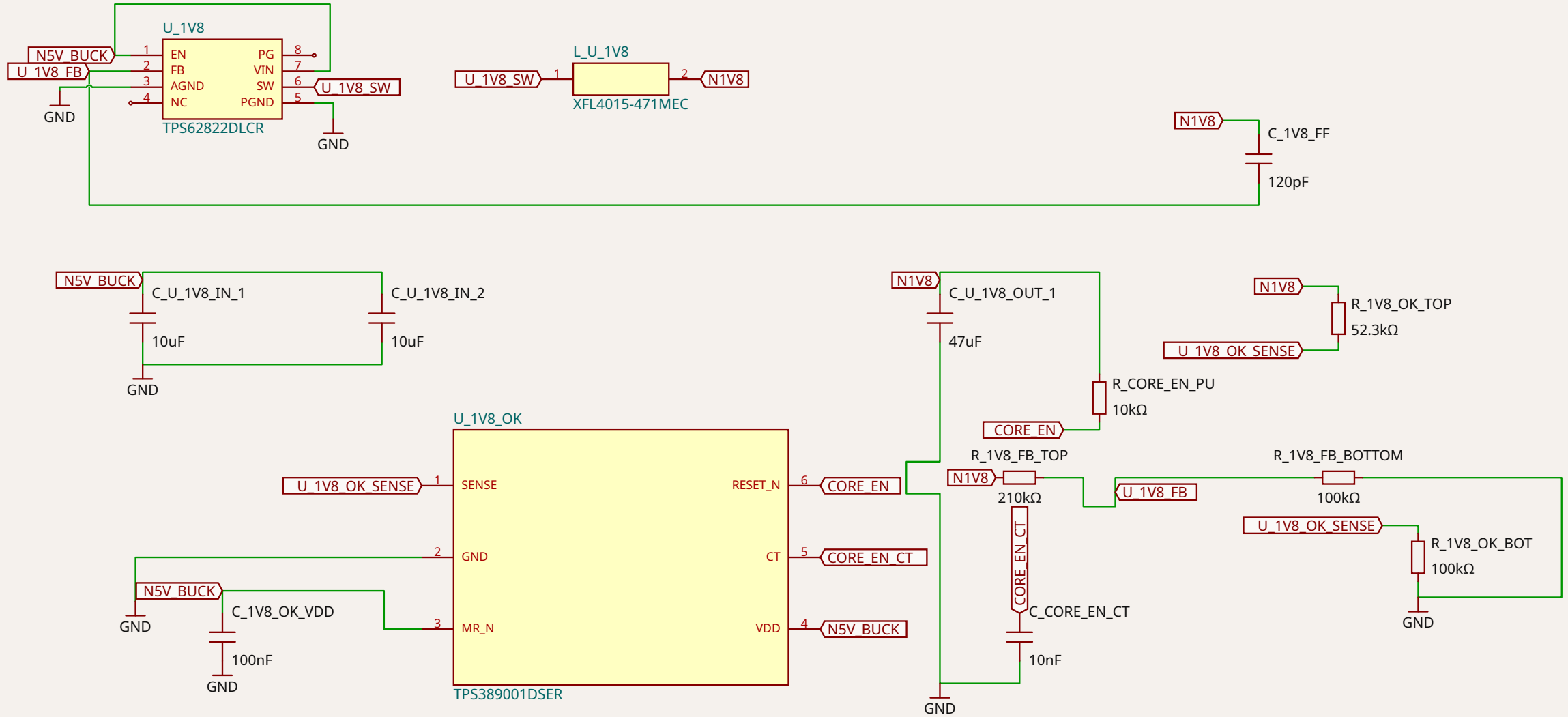
# Crow USB Carrier v1 — PROTOTYPE ONLY

## DIGITAL POWER 3V3X

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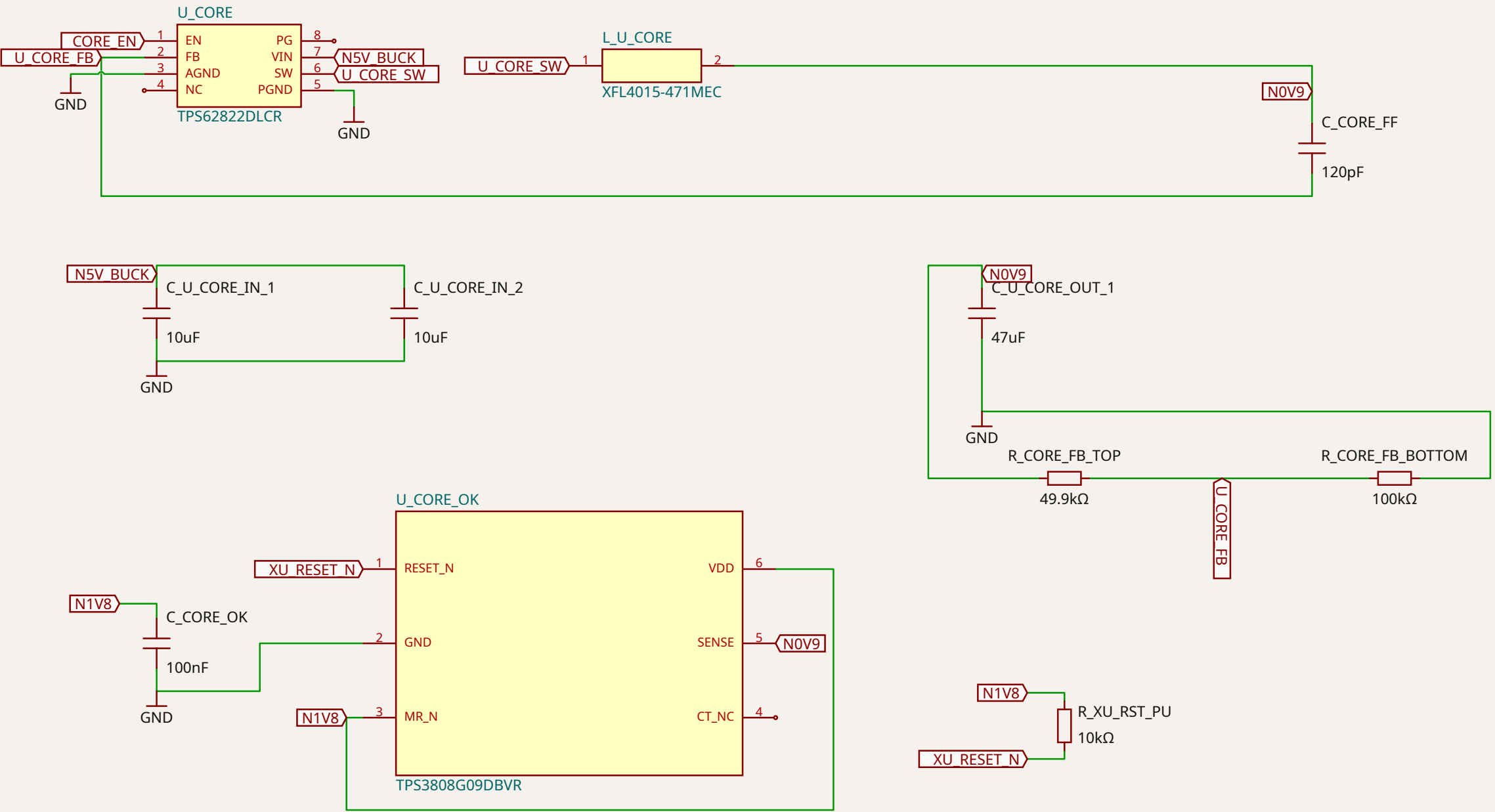


## DIGITAL POWER 1V8



# Crow USB Carrier v1 — PROTOTYPE ONLY

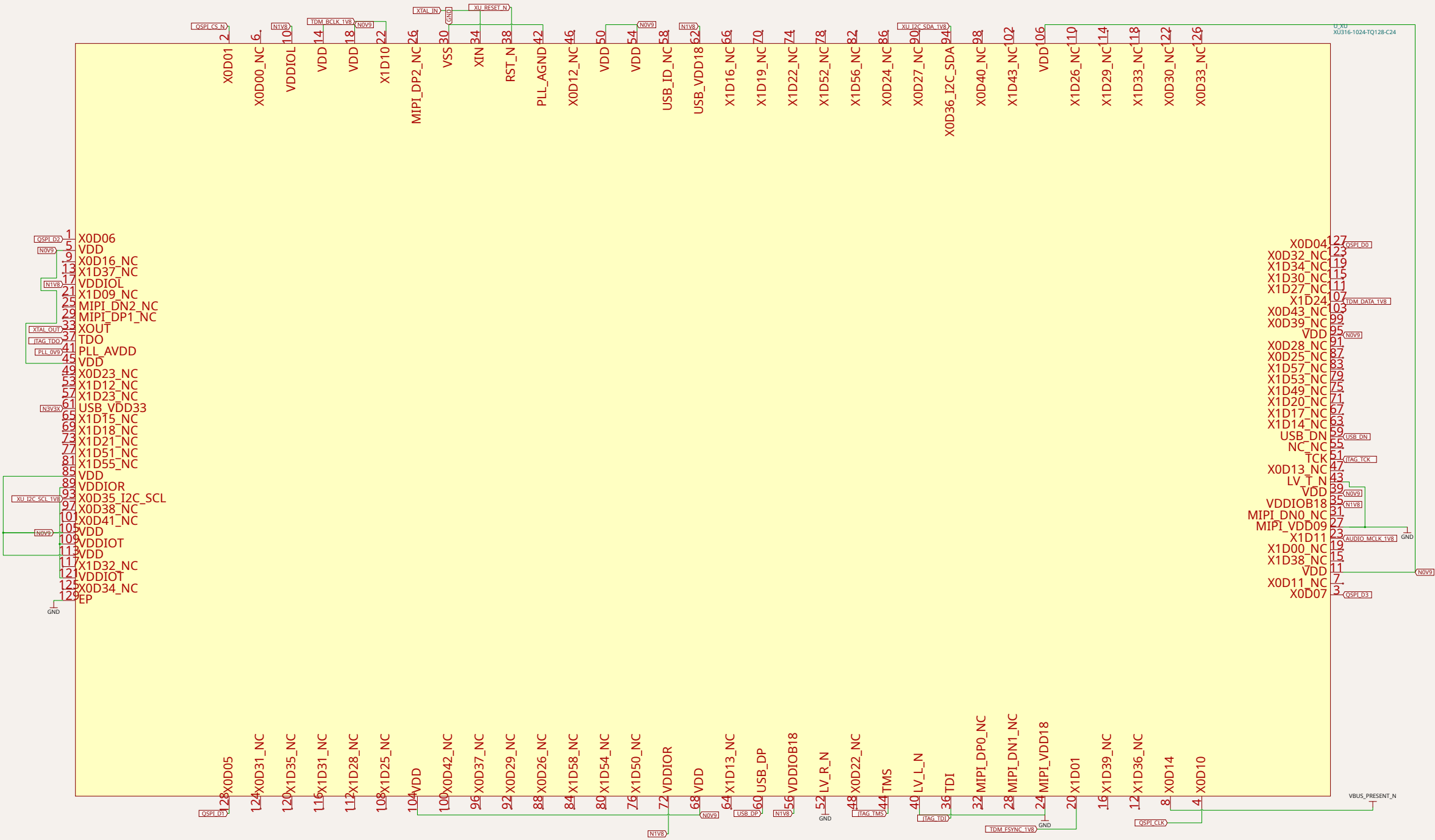
## DIGITAL POWER CORE



# Crow USB Carrier v1 — PROTOTYPE ONLY

## XMOS CORE

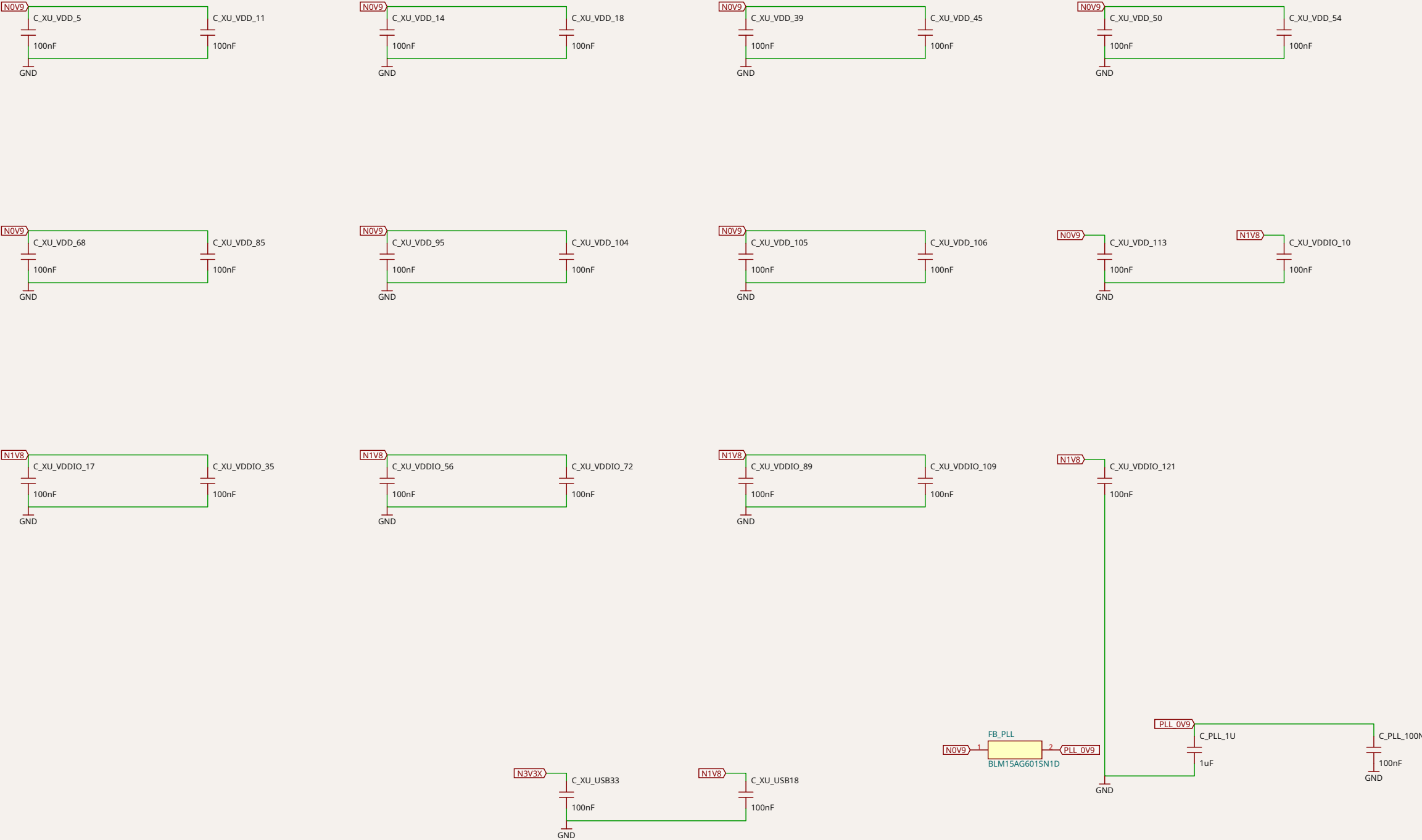
Page 30 of 45 • exact circuit.json SHA-256 580ac31b0de8d376... • 1 components • landscape page fit



# Crow USB Carrier v1 — PROTOTYPE ONLY

## XMOS DECOUPLING

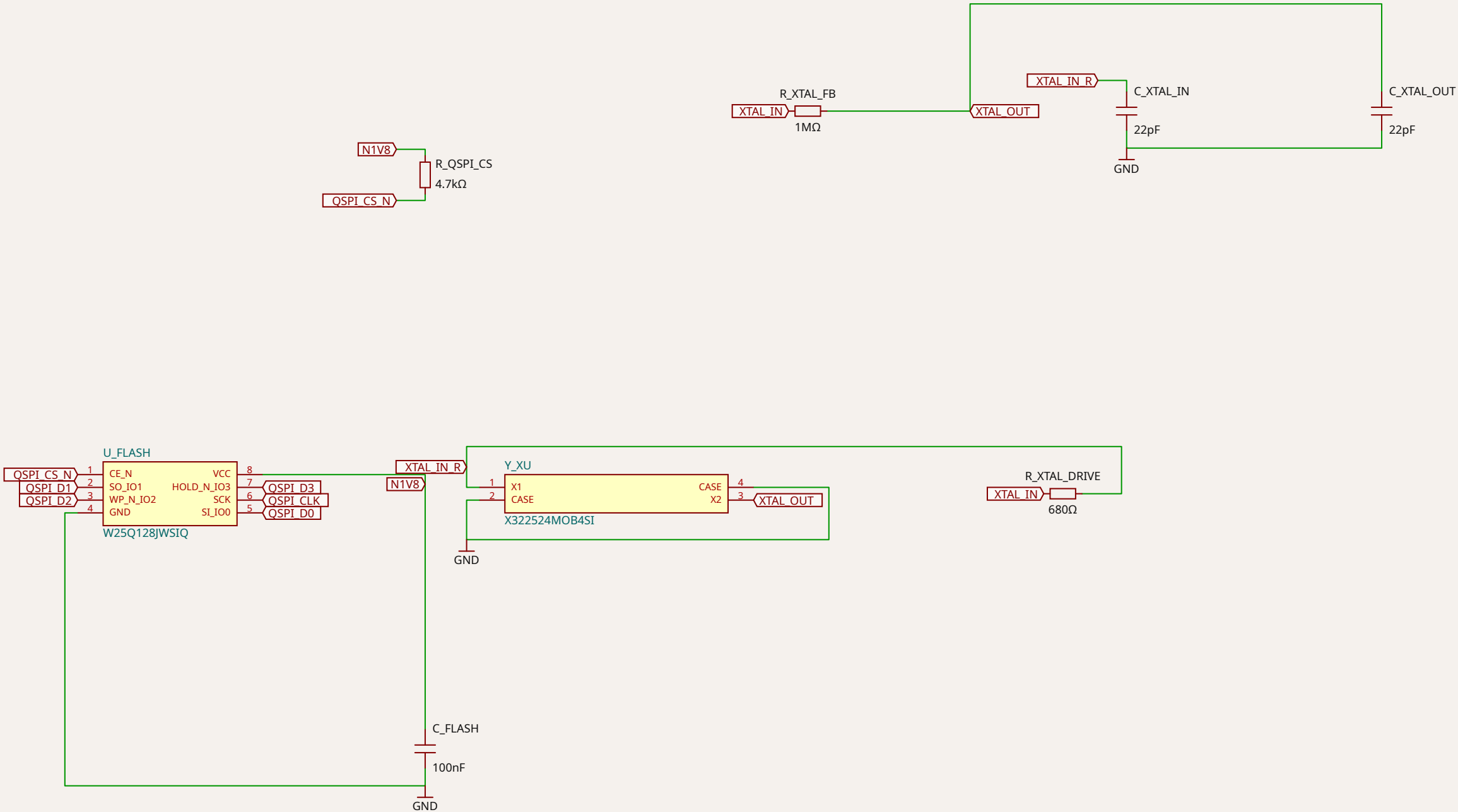
Page 31 of 45 • exact circuit.json SHA-256 580ac31b0de8d376... • 28 components • landscape page fit



# Crow USB Carrier v1 — PROTOTYPE ONLY

## FLASH CLOCK

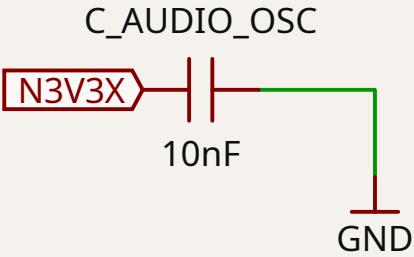
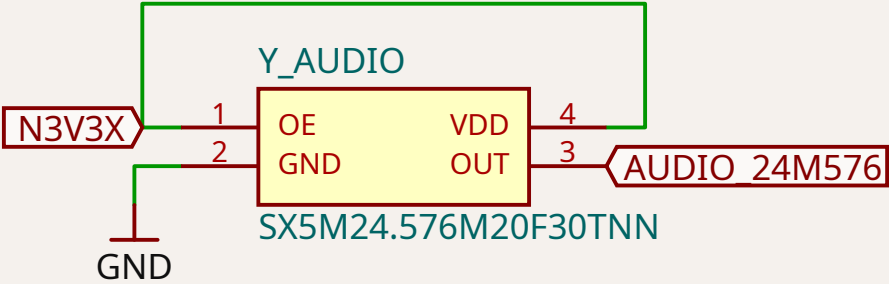
Page 32 of 45 • exact circuit.json SHA-256 580ac31b0de8d376... • 8 components • landscape page fit



# Crow USB Carrier v1 — PROTOTYPE ONLY

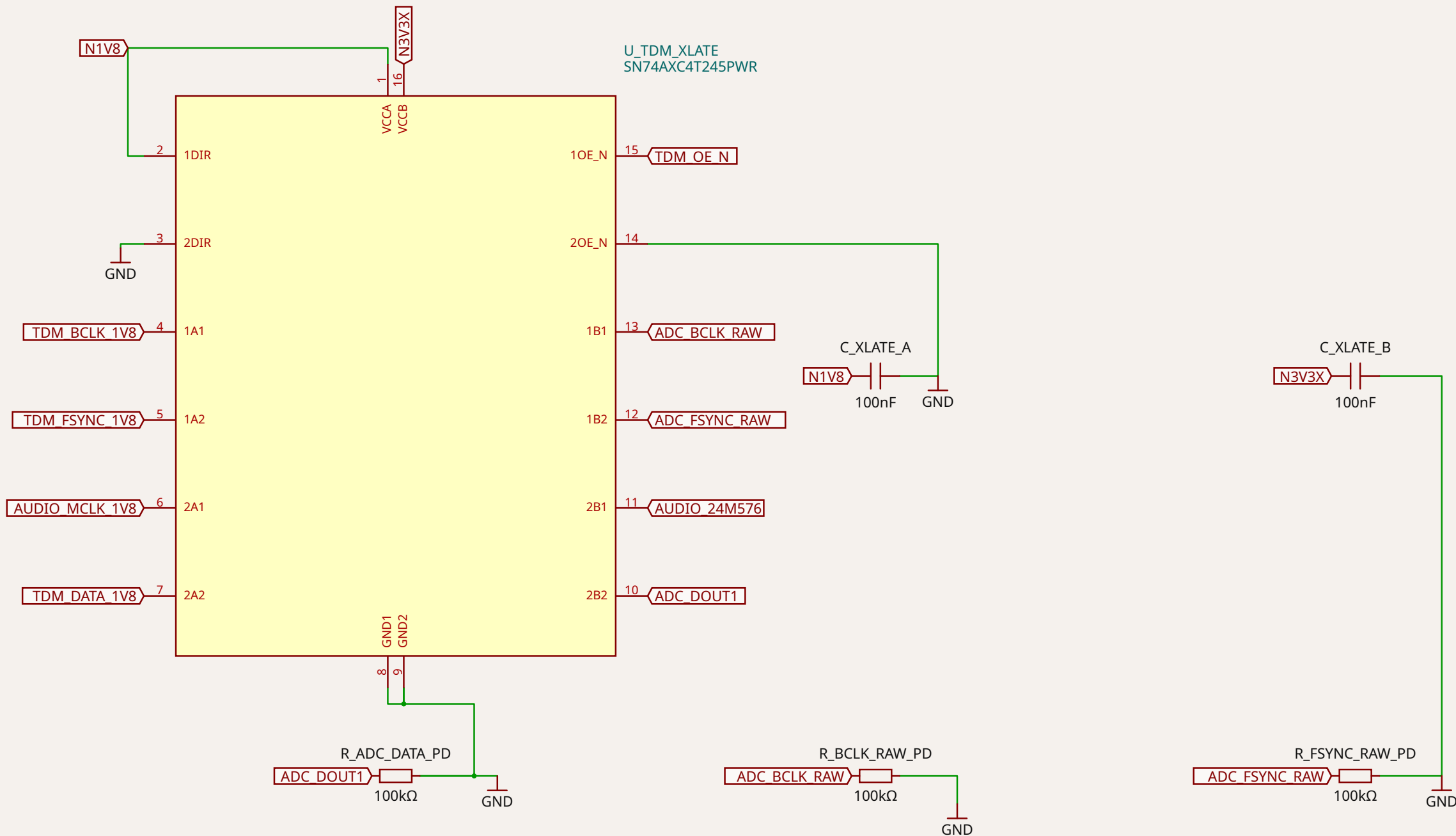
## AUDIO OSCILLATOR

Page 33 of 45 • exact circuit.json SHA-256 580ac31b0de8d376... • 2 components • landscape page fit



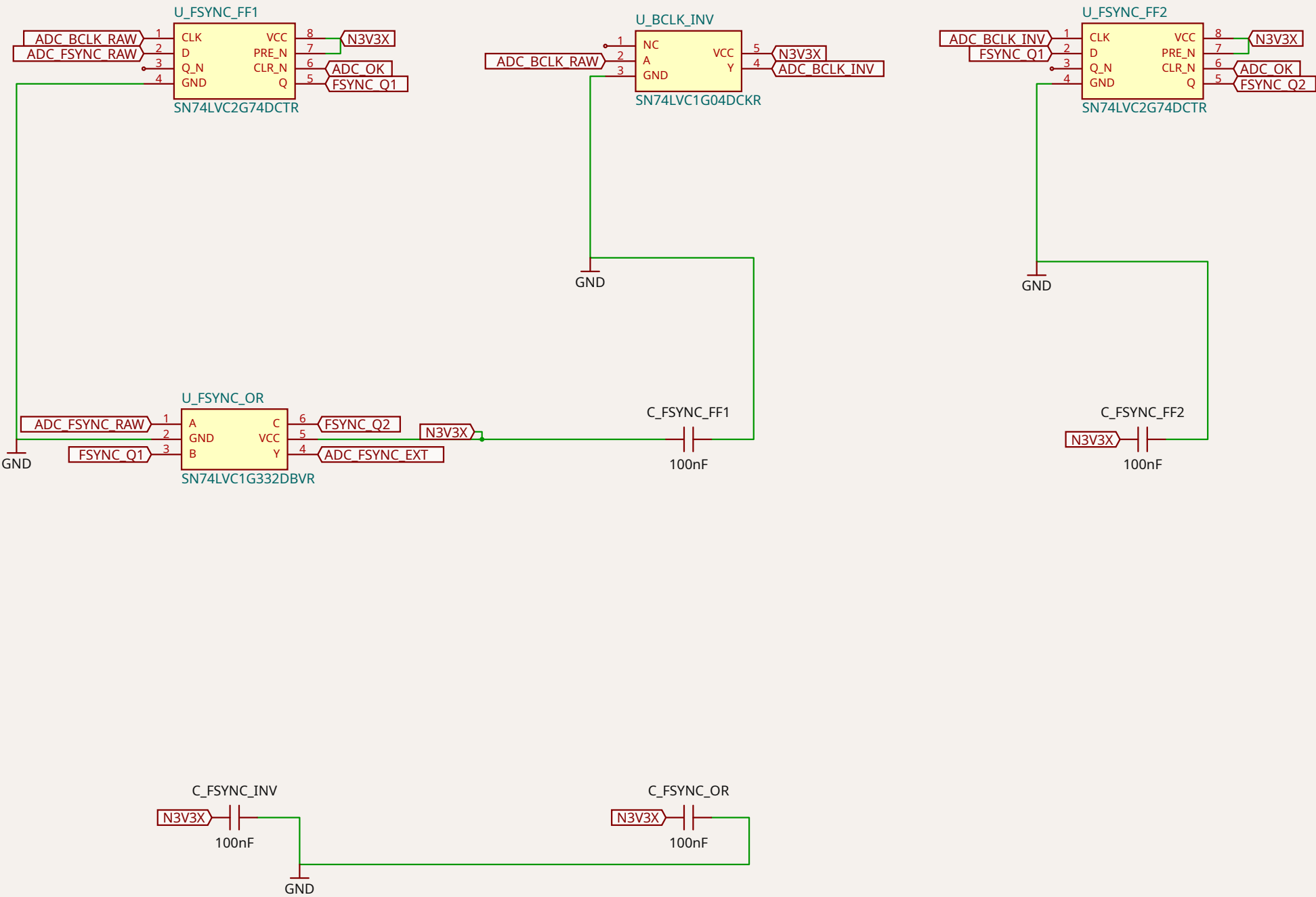
# Crow USB Carrier v1 — PROTOTYPE ONLY

## TDM TRANSLATION



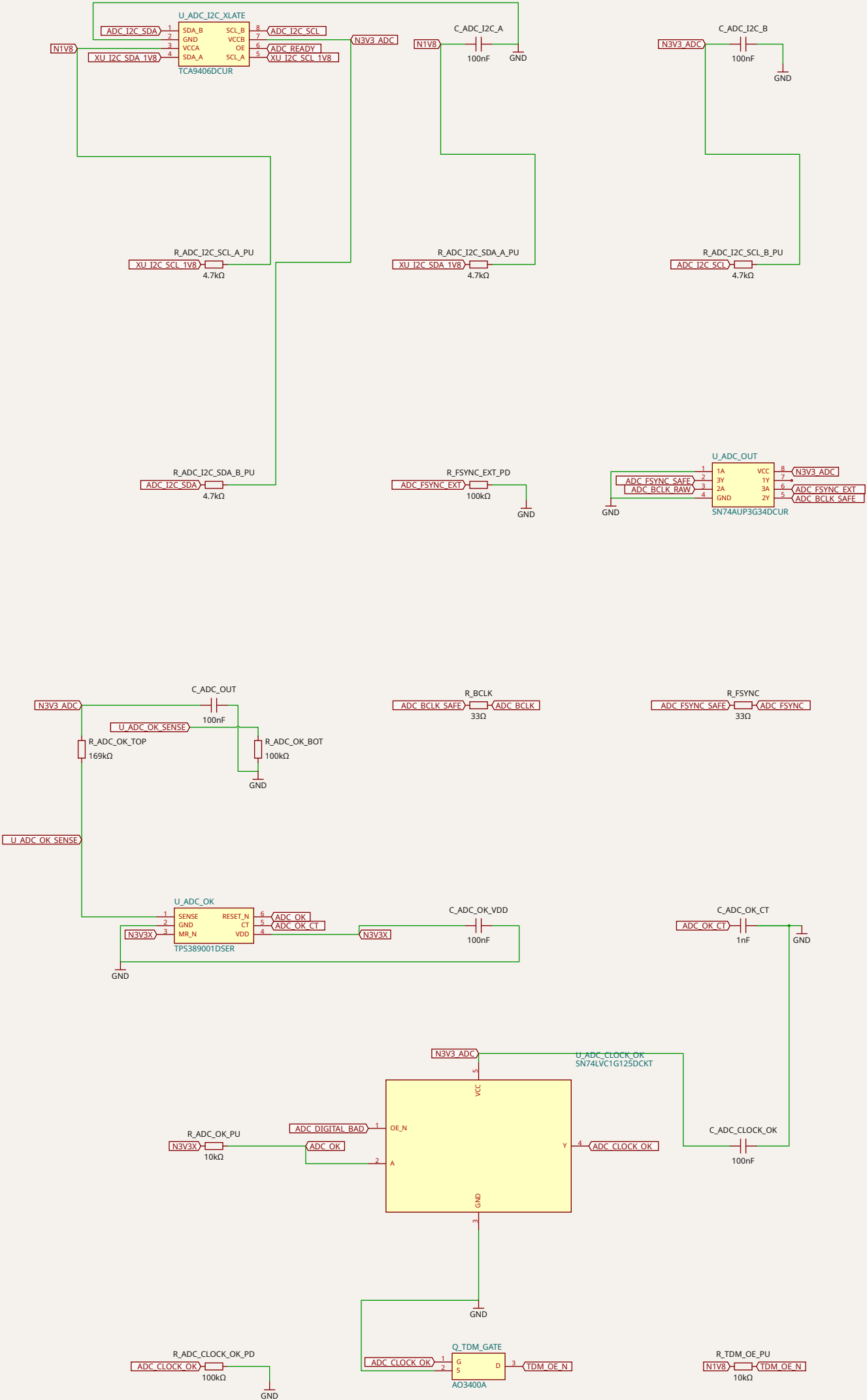
# Crow USB Carrier v1 — PROTOTYPE ONLY

## FSYNC SHAPING



# Crow USB Carrier v1 — PROTOTYPE ONLY

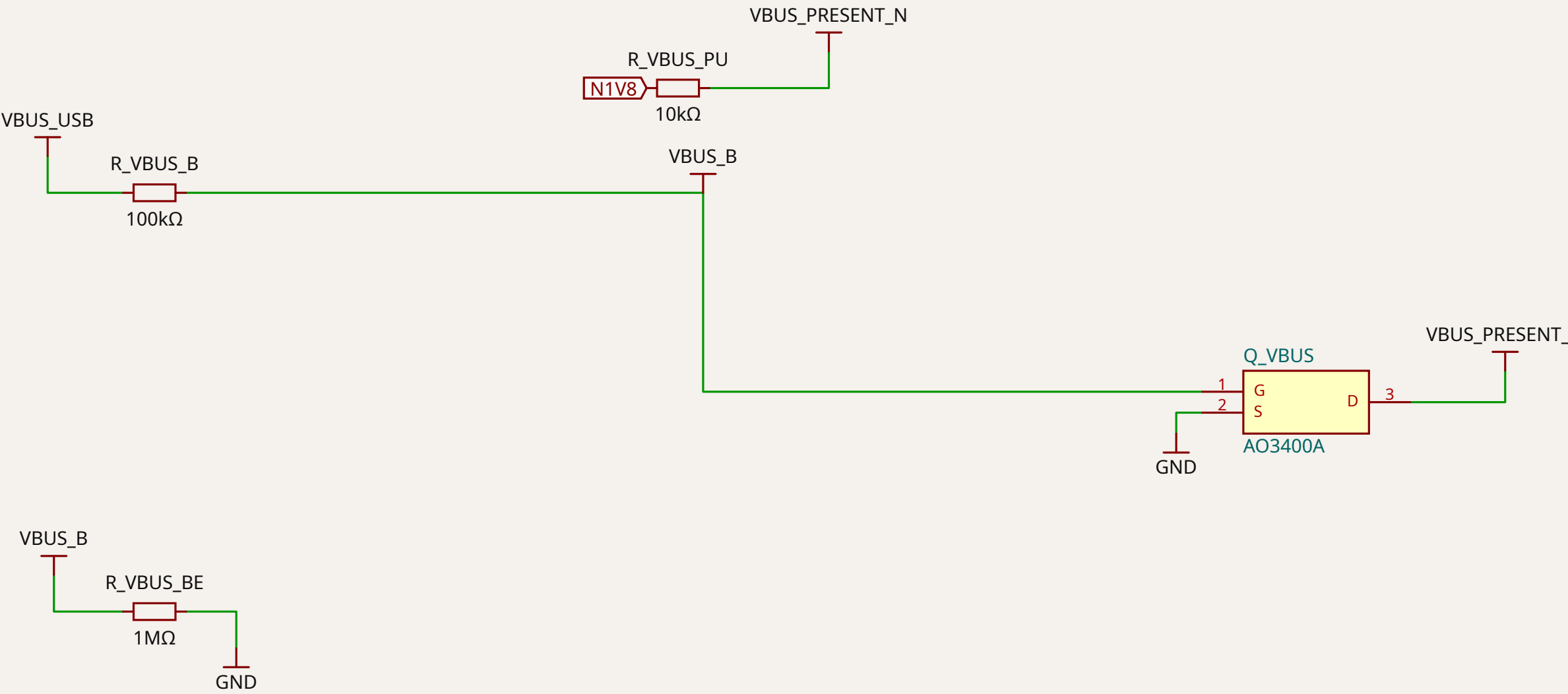
## ADC CLOCK CONTROL



# Crow USB Carrier v1 — PROTOTYPE ONLY

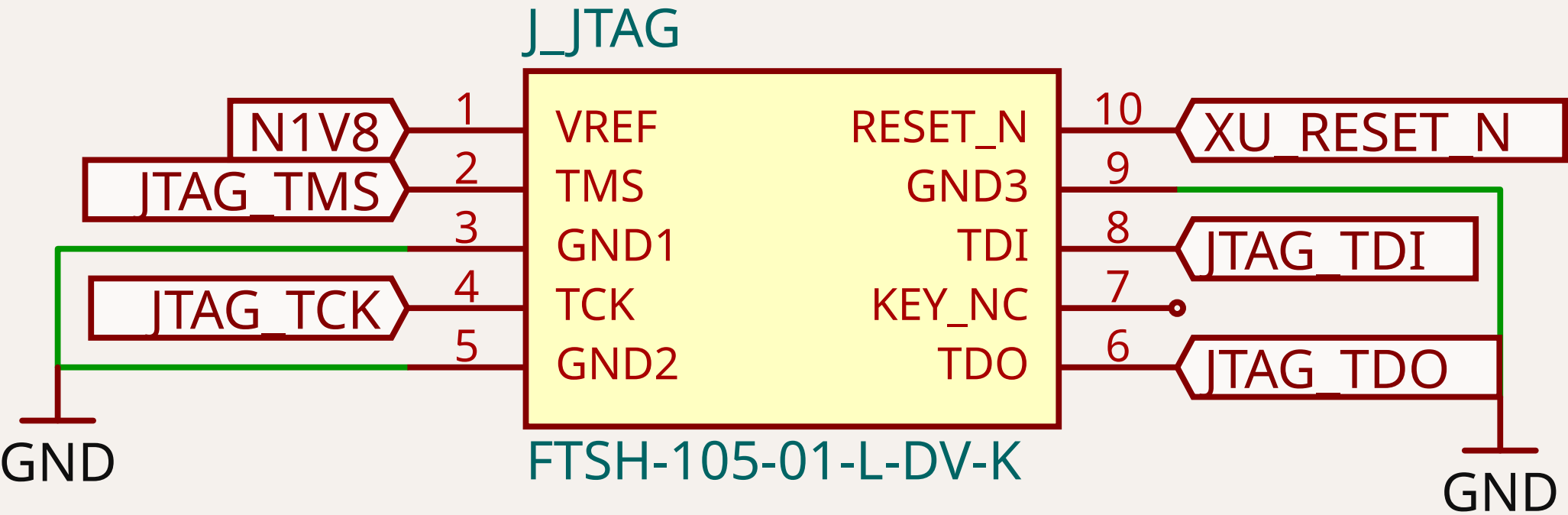
## USB LOGIC

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Crow USB Carrier v1 — PROTOTYPE ONLY  
DEBUG

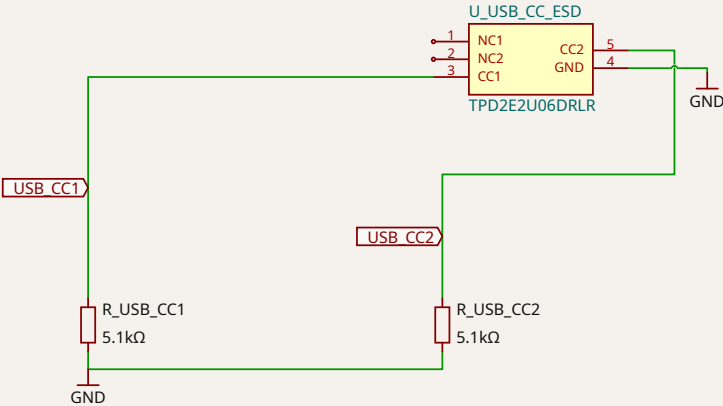
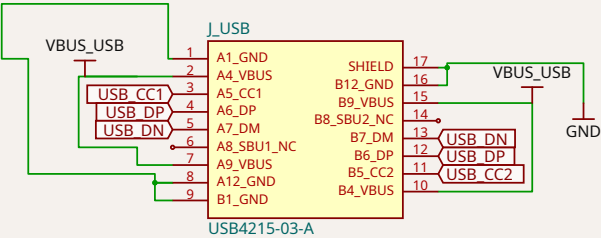
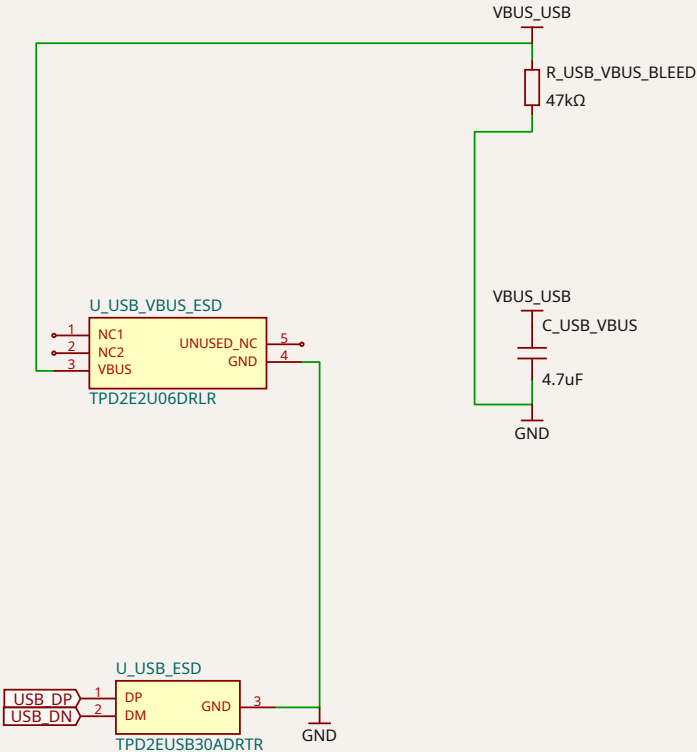
Page 38 of 45 • exact circuit.json SHA-256 580ac31b0de8d376... • 1 components • landscape page fit



# Crow USB Carrier v1 — PROTOTYPE ONLY

## USB FRONTEND

Page 39 of 45 • exact circuit.json SHA-256 580ac31b0de8d376... • 8 components • portrait page fit



# Semantic package-pin assignment appendix

Overview: XMOS CORE — source sheet 30 / schematic\_sheet\_29

Circuit JSON SHA-256: 580ac31b0de8d376d888caa1a342ae43c0636fbd13c74048392173acf718cb6d

Native netlist SHA-256: a40b45c27b2169ed4f3dd43d72a1f4f7112d829985f98918a8855ed76735e9bd

Pin tuples SHA-256: 0f84b3416445350b3c6cc18c9ea49cda78170d5e74a881ceb1d41244ea4b2e29

## 129/129 source pins verified against the fresh native netlist

- 4 audio / TDM
- 2 crystal
- 5 debug / reset
- 6 GND / ground
- 1 GPIO / VBUS sense
- 2 I2C
- 73 NC
- 26 power rail
- 6 QSPI flash
- 2 unused MIPI supply — grounded per XMOS §14
- 2 USB

# Crow USB Carrier v1 — PROTOTYPE ONLY

## XMOS PIN INDEX — U\_XU

### U\_XU package pins 1-26 of 129 — semantic assignment, not a coordinate crop

| Pin | Package pin name | Net / NC       | Function                                   |
|-----|------------------|----------------|--------------------------------------------|
| 1   | X0D06            | QSPI_D2        | QSPI flash                                 |
| 2   | X0D01            | QSPI_CS_N      | QSPI flash                                 |
| 3   | X0D07            | QSPI_D3        | QSPI flash                                 |
| 4   | X0D10            | QSPI_CLK       | QSPI flash                                 |
| 5   | VDD              | N0V9           | power rail                                 |
| 6   | X0D00_NC         | NC             | NC                                         |
| 7   | X0D11_NC         | NC             | NC                                         |
| 8   | X0D14            | VBUS_PRESENT_N | GPIO / VBUS sense                          |
| 9   | X0D16_NC         | NC             | NC                                         |
| 10  | VDDIOL           | N1V8           | power rail                                 |
| 11  | VDD              | N0V9           | power rail                                 |
| 12  | X1D36_NC         | NC             | NC                                         |
| 13  | X1D37_NC         | NC             | NC                                         |
| 14  | VDD              | N0V9           | power rail                                 |
| 15  | X1D38_NC         | NC             | NC                                         |
| 16  | X1D39_NC         | NC             | NC                                         |
| 17  | VDDIOL           | N1V8           | power rail                                 |
| 18  | VDD              | N0V9           | power rail                                 |
| 19  | X1D00_NC         | NC             | NC                                         |
| 20  | X1D01            | TDM_FSYNC_1V8  | audio / TDM                                |
| 21  | X1D09_NC         | NC             | NC                                         |
| 22  | X1D10            | TDM_BCLK_1V8   | audio / TDM                                |
| 23  | X1D11            | AUDIO_MCLK_1V8 | audio / TDM                                |
| 24  | MIPI_VDD18       | GND            | unused MIPI supply — grounded per XMOS §14 |
| 25  | MIPI_DN2_NC      | NC             | NC                                         |
| 26  | MIPI_DP2_NC      | NC             | NC                                         |

# Crow USB Carrier v1 — PROTOTYPE ONLY

## XMOS PIN INDEX — U\_XU

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### U\_XU package pins 27–52 of 129 — semantic assignment, not a coordinate crop

| Pin | Package pin name | Net / NC   | Function                                   |
|-----|------------------|------------|--------------------------------------------|
| 27  | MIPI_VDD09       | GND        | unused MIPI supply — grounded per XMOS §14 |
| 28  | MIPI_DN1_NC      | NC         | NC                                         |
| 29  | MIPI_DP1_NC      | NC         | NC                                         |
| 30  | VSS              | GND        | GND / ground                               |
| 31  | MIPI_DN0_NC      | NC         | NC                                         |
| 32  | MIPI_DP0_NC      | NC         | NC                                         |
| 33  | XOUT             | XTAL_OUT   | crystal                                    |
| 34  | XIN              | XTAL_IN    | crystal                                    |
| 35  | VDDI0B18         | N1V8       | power rail                                 |
| 36  | TDI              | JTAG_TDI   | debug / reset                              |
| 37  | TD0              | JTAG_TD0   | debug / reset                              |
| 38  | RST_N            | XU_RESET_N | debug / reset                              |
| 39  | VDD              | N0V9       | power rail                                 |
| 40  | LV_L_N           | GND        | GND / ground                               |
| 41  | PLL_AVDD         | PLL_0V9    | power rail                                 |
| 42  | PLL_AGND         | GND        | GND / ground                               |
| 43  | LV_T_N           | GND        | GND / ground                               |
| 44  | TMS              | JTAG_TMS   | debug / reset                              |
| 45  | VDD              | N0V9       | power rail                                 |
| 46  | X0D12_NC         | NC         | NC                                         |
| 47  | X0D13_NC         | NC         | NC                                         |
| 48  | X0D22_NC         | NC         | NC                                         |
| 49  | X0D23_NC         | NC         | NC                                         |
| 50  | VDD              | N0V9       | power rail                                 |
| 51  | TCK              | JTAG_TCK   | debug / reset                              |
| 52  | LV_R_N           | GND        | GND / ground                               |

# Crow USB Carrier v1 — PROTOTYPE ONLY

## XMOS PIN INDEX — U\_XU

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### U\_XU package pins 53–78 of 129 — semantic assignment, not a coordinate crop

| Pin | Package pin name | Net / NC | Function   |
|-----|------------------|----------|------------|
| 53  | X1D12_NC         | NC       | NC         |
| 54  | VDD              | N0V9     | power rail |
| 55  | NC_NC            | NC       | NC         |
| 56  | VDDI0B18         | N1V8     | power rail |
| 57  | X1D23_NC         | NC       | NC         |
| 58  | USB_ID_NC        | NC       | NC         |
| 59  | USB_DN           | USB_DN   | USB        |
| 60  | USB_DP           | USB_DP   | USB        |
| 61  | USB_VDD33        | N3V3X    | power rail |
| 62  | USB_VDD18        | N1V8     | power rail |
| 63  | X1D14_NC         | NC       | NC         |
| 64  | X1D13_NC         | NC       | NC         |
| 65  | X1D15_NC         | NC       | NC         |
| 66  | X1D16_NC         | NC       | NC         |
| 67  | X1D17_NC         | NC       | NC         |
| 68  | VDD              | N0V9     | power rail |
| 69  | X1D18_NC         | NC       | NC         |
| 70  | X1D19_NC         | NC       | NC         |
| 71  | X1D20_NC         | NC       | NC         |
| 72  | VDDI0R           | N1V8     | power rail |
| 73  | X1D21_NC         | NC       | NC         |
| 74  | X1D22_NC         | NC       | NC         |
| 75  | X1D49_NC         | NC       | NC         |
| 76  | X1D50_NC         | NC       | NC         |
| 77  | X1D51_NC         | NC       | NC         |
| 78  | X1D52_NC         | NC       | NC         |

# Crow USB Carrier v1 — PROTOTYPE ONLY

## XMOS PIN INDEX — U\_XU

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### U\_XU package pins 79–104 of 129 — semantic assignment, not a coordinate crop

| Pin | Package pin name | Net / NC       | Function   |
|-----|------------------|----------------|------------|
| 79  | X1D53_NC         | NC             | NC         |
| 80  | X1D54_NC         | NC             | NC         |
| 81  | X1D55_NC         | NC             | NC         |
| 82  | X1D56_NC         | NC             | NC         |
| 83  | X1D57_NC         | NC             | NC         |
| 84  | X1D58_NC         | NC             | NC         |
| 85  | VDD              | N0V9           | power rail |
| 86  | X0D24_NC         | NC             | NC         |
| 87  | X0D25_NC         | NC             | NC         |
| 88  | X0D26_NC         | NC             | NC         |
| 89  | VDDI0R           | N1V8           | power rail |
| 90  | X0D27_NC         | NC             | NC         |
| 91  | X0D28_NC         | NC             | NC         |
| 92  | X0D29_NC         | NC             | NC         |
| 93  | X0D35_I2C_SCL    | XU_I2C_SCL_1V8 | I2C        |
| 94  | X0D36_I2C_SDA    | XU_I2C_SDA_1V8 | I2C        |
| 95  | VDD              | N0V9           | power rail |
| 96  | X0D37_NC         | NC             | NC         |
| 97  | X0D38_NC         | NC             | NC         |
| 98  | X0D40_NC         | NC             | NC         |
| 99  | X0D39_NC         | NC             | NC         |
| 100 | X0D42_NC         | NC             | NC         |
| 101 | X0D41_NC         | NC             | NC         |
| 102 | X1D43_NC         | NC             | NC         |
| 103 | X0D43_NC         | NC             | NC         |
| 104 | VDD              | N0V9           | power rail |

# Crow USB Carrier v1 — PROTOTYPE ONLY

## XMOS PIN INDEX — U\_XU

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### U\_XU package pins 105–129 of 129 — semantic assignment, not a coordinate crop

| Pin | Package pin name | Net / NC     | Function     |
|-----|------------------|--------------|--------------|
| 105 | VDD              | N0V9         | power rail   |
| 106 | VDD              | N0V9         | power rail   |
| 107 | X1D24            | TDM_DATA_1V8 | audio / TDM  |
| 108 | X1D25_NC         | NC           | NC           |
| 109 | VDDI0T           | N1V8         | power rail   |
| 110 | X1D26_NC         | NC           | NC           |
| 111 | X1D27_NC         | NC           | NC           |
| 112 | X1D28_NC         | NC           | NC           |
| 113 | VDD              | N0V9         | power rail   |
| 114 | X1D29_NC         | NC           | NC           |
| 115 | X1D30_NC         | NC           | NC           |
| 116 | X1D31_NC         | NC           | NC           |
| 117 | X1D32_NC         | NC           | NC           |
| 118 | X1D33_NC         | NC           | NC           |
| 119 | X1D34_NC         | NC           | NC           |
| 120 | X1D35_NC         | NC           | NC           |
| 121 | VDDI0T           | N1V8         | power rail   |
| 122 | X0D30_NC         | NC           | NC           |
| 123 | X0D32_NC         | NC           | NC           |
| 124 | X0D31_NC         | NC           | NC           |
| 125 | X0D34_NC         | NC           | NC           |
| 126 | X0D33_NC         | NC           | NC           |
| 127 | X0D04            | QSPI_D0      | QSPI flash   |
| 128 | X0D05            | QSPI_D1      | QSPI flash   |
| 129 | EP               | GND          | GND / ground |